Two-Sided Cumulative Dominance in Organizational Hierarchies

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Abstract

This paper compares finite organizational hierarchies using the two anonymous margins of the effective authority relation. The burden profile records each position's number of direct and indirect supervisors, while the reach profile records its number of direct and indirect subordinates. Absolute two-sided dominance compares both profiles in their original units, while proportional two-sided dominance compares proportional changes in inclusive burden and reach. Both product preorders admit exact robust-unanimity representations: a comparison holds precisely when it is supported by every evaluation obtained by applying separate nondecreasing convex functions to burden and reach and assigning positive relative weights to the two profiles. The resulting comparisons identify rankings that are robust to these unspecified numerical judgments and cases in which additional organizational preferences are required. For absolute dominance, every index within a broad position-additive class that preserves all product comparisons admits an equivalent representation of this form. A tree–diamond example shows how conflicting marginal rankings and burden–reach tradeoffs produce incomparability. Both relations satisfy Subordination Removal and preserve every weak and strict sequential-removal comparison, while admitting additional comparisons. For comparisons across population sizes, the paper develops per-position and aggregate extensions and studies a selected extension that measures each position's burden and reach relative to what is possible at that population size. Organizational growth affects these interpretations differently: vertical attachment always raises aggregate hierarchy, but need not raise hierarchy per position or relative to the authority relations possible in the larger population.

Keywords: organizational hierarchy; incomplete ordering; majorization; robust unanimity; directed acyclic graphs; variable-population comparison.

JEL classifications: C02; D23; L22.

1 Introduction

Organizational hierarchy concerns both sides of authority. Some positions exercise authority over many others, directly or through chains of command. Other positions are subject to many supervisors, again directly or indirectly. A hierarchy therefore differentiates organizational positions along two dimensions: authority over subordinates and subordination to supervisors.

These dimensions are familiar in the theory and empirical analysis of organizations. Hierarchical models emphasize limits on supervision, loss of control across organizational levels, optimal spans of control, scale-of-operations effects, and the allocation of knowledge and decision making (Williamson, 1967; Calvo and Wellisz, 1978; Keren and Levhari, 1979; Rosen, 1982; Garicano, 2000). Empirical work studies organizational layers, reporting relations, managerial spans, delegation, and reorganization within firms (Rajan and Wulf, 2006; Caliendo, Monte, and Rossi-Hansberg, 2015; Ewens and Giroud, 2025). This literature

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shows that hierarchy cannot generally be reduced to the number of formal levels alone. Authority may be concentrated or dispersed within a given layer structure, and positions at the same formal level may differ substantially in the organizational domains over which they exercise or bear authority.

A measure based on only one side of authority can give an incomplete or perspective-dependent account. An organization may concentrate multiple lines of supervision on one deeply subordinated position while dividing subordinate reach among several intermediate supervisors. Another may concentrate intermediate authority while spreading deep subjection across several positions. Ranking one above the other requires prioritizing one side of authority or selecting a tradeoff between burden and reach.

The network literature has proposed scalar measures of hierarchy based on degree, path structure, flow, random walks, and heterogeneity in reach (Trusina et al., 2004; Luo and Magee, 2011; Mones, Vicsek, and Vicsek, 2012; Corominas-Murtra et al., 2013; Czégel and Palla, 2015). Such measures summarize particular structural features. The present paper instead compares the complete anonymous distributions of subordinate reach and supervisory burden without selecting one scalar feature or one compensatory aggregation rule.

The paper proposes *two-sided cumulative dominance*. An organizational hierarchy is represented by a finite directed acyclic graph. Its strict reachability relation records all direct and indirect superior–subordinate pairs. The effective-burden profile records each position's number of direct and indirect supervisors, while the effective-reach profile records its number of direct and indirect subordinates. These profiles are the two anonymous margins of the same effective authority relation.

The paper studies two scales for comparing the profiles. The absolute relation accumulates numbers of effective authority relations. The proportional relation accumulates logarithms of inclusive burden and reach, where an inclusive domain contains a position together with its direct and indirect supervisors or subordinates. The absolute scale assigns equal weight to equal numerical differences. The proportional scale assigns equal weight to equal proportional changes in inclusive domains and discounts additional relations at positions whose domains are already large.

One hierarchy two-sidedly dominates another when both anonymous profiles cumulatively dominate their counterparts. Equivalently, at every rank cutoff, the first hierarchy has at least as much effective subjection among its most subordinated positions and at least as much effective authority among its most authoritative positions. The criterion is incomplete. When the two profiles support opposite rankings, or when their cumulative comparisons cross, the hierarchies remain incomparable. For a given reachability relation, the comparison is implemented by sorting the burden and reach profiles and checking their cumulative values, measured on the relevant scale, at every cutoff.

This incompleteness admits a robust-unanimity representation. Consider separable marginal evaluations that are nondecreasing and convex in burden and reach measured on the relevant scale: raw counts for absolute dominance and logarithmic inclusive counts for proportional dominance. Monotonicity treats additional authority or subjection as hierarchy enhancing. Convexity makes the evaluation responsive to concentration: holding the total on the relevant scale fixed, shifting burden or reach toward positions that already have higher values weakly raises the evaluation. Separability leaves aside judgments about the association between burden and reach across positions. In particular, the evaluation does not depend on whether positions with high burden also have high reach. Allowing every positive burden and reach weight leaves their relative importance unrestricted.

For either scale, two-sided cumulative dominance is exactly the unanimous relation generated by this numerical family. The representation identifies the normative content of the product preorder. Incomparability records the absence of a ranking robust to the full admissible family, rather than an unexplained refusal to aggregate. A selected weighted evaluation may rank a pair on which the family does not agree. Lexicographic and nonseparable rules may likewise rank some pairs left unresolved by the product preorder, but only by introducing judgments about priority or burden–reach association.

The criterion also has a direct organizational interpretation. Every effective authority relation contributes one unit of reach to a supervisory position and one unit of burden to a subordinate position. Under absolute

dominance, the terminal cutoff compares the volume of these relations, while the smaller cutoffs compare their distribution across burden and reach. Under proportional dominance, burden and reach are measured relative to the size of each position's existing inclusive domain. An additional supervisor or subordinate, therefore, has greater significance when that domain is initially small. Under either scale, a dominating hierarchy has at least as much hierarchical stratification from both perspectives: supervisory burden and subordinate reach.

The representation applies the standard characterization of weak majorization by nondecreasing convex functions (Marshall, Olkin, and Arnold, 2011, Theorem 4.B.2) separately to burden and reach.

The analysis yields four principal results. First, the fixed-size unanimity theorem characterizes the robust ordinal comparison of organizational hierarchies when the marginal evaluations and relative burden–reach weights are left unspecified. Two-sided cumulative dominance ranks one hierarchy above another exactly when every admissible evaluation does so. Equivalently, a hierarchical preorder satisfies Robust Marginal Unanimity exactly when it preserves every weak and strict comparison made by the corresponding product preorder. For absolute dominance, a stronger maximality theorem considers indices formed by summing a common position-level evaluation of burden and reach. If the same evaluation preserves every absolute product comparison across all finite population sizes, the resulting index admits an equivalent representation that is additively separable in burden and reach, with nondecreasing, discretely convex components. Equivalent evaluations assign the same aggregate index value to every hierarchy.

Second, the tree–diamond comparison in Section 5 shows that burden and reach can rank simple hierarchies in opposite directions. Different admissible burden–reach weights then produce opposite scalar rankings, so neither strict verdict survives the full numerical family. Incomparability is the robust conclusion unless an application supplies an organizational basis for giving one profile priority or selecting a burden–reach tradeoff.

Third, both product relations satisfy Subordination Removal: deleting a direct reporting relation, while adding only the bypass edges needed to preserve the remaining chains of command, cannot increase hierarchy and strictly lowers it when an effective authority relation is lost. The product relations also preserve every weak and strict comparison generated by a sequence of such removals. The converse fails: they admit comparisons that cannot be implemented by removal sequences. These results are established in Sections 6.4 and 6.5.

Fourth, the paper separates two choices in comparisons across population sizes: whether burden and reach are measured in absolute or proportional terms, and how organizational size enters the comparison. The first cross-size extension compares hierarchy per position by replicating organizations to a common population. It is uniquely determined by fixed-size agreement and replication invariance. Vertical attachment raises hierarchy under this interpretation when the appended component is at least as hierarchical as the base, but not necessarily without that condition. The second extension compares aggregate hierarchy by adjoining isolated positions to the smaller organization. It is uniquely determined by fixed-size agreement and neutrality toward isolated positions, and every admissible vertical attachment raises hierarchy. The third extension compares each position's authority and subjection with what would be possible in an organization of the same size. Under this interpretation, adding positions can lower measured hierarchy even when it creates new authority relations, because the range of possible authority relations expands as well. The three extensions admit corresponding numerical representations.

1.1 Relation to earlier work

The paper builds most directly on three related studies. Carbonell-Nicolau (2025) develops an axiomatic hierarchy preorder for trees based on Subordination Removal. Carbonell-Nicolau (2026b) studies the sequential implementation of that operation on DAGs with multiple supervisors. Carbonell-Nicolau (2026a) studies complete scalar depth–span evaluations and distinguishes average, total, and concentration interpretations through their population normalizations, replication properties, and responses to structural extension.

The present paper inherits the effective authority relation and its burden and reach coordinates from these studies. It also uses Subordination Removal, replication, and structural extension as common consistency benchmarks and uses the depth–span architecture to formulate numerical representations. These concepts provide a common domain and common tests through which the new product preorders can be compared with the earlier operational and numerical approaches.

Specific to this paper are the absolute and proportional product preorders and the results developed for them. Unlike the earlier operational preorders and complete scalar indices, the present relations are incomplete distributional orders constructed from the two anonymous margins of effective authority. The paper establishes their robust-unanimity representations, proves that they preserve every sequential-removal comparison while admitting additional comparisons, and develops three cross-size extensions with distinct responses to structural attachment. For absolute dominance, it also shows that every compatible index within the stated all-size, population-independent, position-additive class admits an equivalent representation with nondecreasing, discretely convex burden and reach components.

The tree and diamond in Section 5 receive opposite one-sided rankings and are therefore left incomparable by the product criterion. Some product comparisons cannot be generated by sequential removals. Individual scalar depth–span indices can rank additional pairs by selecting a burden–reach weight or valuing burden–reach association, whereas the product preorders retain only comparisons unanimously supported by the admissible marginal family.

Methodologically, the paper takes an ordinal-measurement approach to organizational hierarchy. Its objective is to compare given structures, not to explain their endogenous formation or estimate reporting relations from data. The organizational setting determines what is being compared and how burden and reach are interpreted, while majorization, order theory, and variable-population comparison provide the mathematical methods.

The paper is related to the economic literature on organizational hierarchies. Early models explain hierarchy through limits on control, supervisory capacity, optimal spans of control, and scale-of-operations effects, under which managerial ability becomes more valuable when it affects more positions in the hierarchy (Williamson, 1967; Calvo and Wellisz, 1978; Keren and Levhari, 1979; Rosen, 1982). Knowledge-based models derive hierarchical organization from the allocation of problem-solving knowledge and communication (Garicano, 2000). Empirical studies document reporting relations, managerial spans, delegation, and organizational layers (Rajan and Wulf, 2006; Caliendo, Monte, and Rossi-Hansberg, 2015; Ewens and Giroud, 2025). The present paper addresses a subsequent measurement question. Given an observed or reconstructed reporting hierarchy, it provides an ordinal method for comparing the distributions of subordinate reach and supervisory burden across positions.

The paper is also related to multiple criteria decision analysis (Greco, Słowiński, and Wallenius, 2025). Organizational hierarchies may be viewed as alternatives evaluated through the potentially conflicting burden and reach profiles. Multiple criteria methods generally combine evaluations across criteria through a preference model representing the relevant decision maker's value system. Robust approaches instead consider a family of admissible preference models and identify comparisons supported throughout that family. The present framework shares this robustness perspective but does not infer the admissible family from elicited preferences. It specifies the family through monotonicity, convexity, separability, and freely varying positive burden–reach weights, and derives direct cumulative-profile conditions for the comparisons supported by every admissible evaluation.

The paper also differs from network measures that summarize hierarchy through one scalar statistic (Trusina et al., 2004; Luo and Magee, 2011; Mones, Vicsek, and Vicsek, 2012; Corominas-Murtra et al., 2013; Czégel and Palla, 2015). Rather than reducing hierarchy to one statistic, the present criterion compares the complete anonymous reach and burden distributions through their cumulative profiles. The companion index paper shows that measures developed for other notions of network hierarchy need not satisfy the organizational monotonicity principles used in this research program (Carbonell-Nicolau, 2026a). This does not make those

measures unsuitable for their original purposes; it reflects a difference in the concepts being measured.

Finally, majorization supplies the mathematical tools used to compare the anonymous profiles and derive the unanimity representation. Its use here is two-sided: one cumulative comparison is applied to burden and another to reach. The contribution lies in identifying the product preorder as the robust core of the admissible marginal-evaluation family and studying its behavior under graph operations and population change.

Sections 2 to 4 develop the fixed-size framework and its interpretation. Sections 5 and 6 establish the principal fixed-size implications. Sections 7 to 10 develop the cross-size theory. Section 11 gives the unanimity and maximality results, while Sections 12 and 13 discuss scope and conclude.

2 Effective Authority, Supervisory Burden, and Subordinate Reach

Fix a finite set V of organizational positions, with $|V| = n$, and let $\mathcal{H}(V)$ denote the set of directed acyclic graphs on V. An element

$$G = (V, E_G) \in \mathcal{H}(V)$$

is an *organizational hierarchy*, where $E_G \subseteq V \times V$ is its edge set. An edge $(i, j) \in E_G$, written $i \to j$, means that position j reports directly to position i. Edge direction therefore runs from supervisor to direct report.

A directed path of positive length from i to j is written $i \rightsquigarrow_G j$. The strict reachability relation is

$$P_G = \{(i, j) \in V \times V : i \rightsquigarrow_G j\}.$$

A pair $(i, j) \in P_G$ is an *effective authority relation*: position i is a direct or indirect supervisor of position j, and position j is a direct or indirect subordinate of position i.

For each $j \in V$, define its ancestor set by

$$\mathrm{Anc}_G(j) = \{i \in V : (i, j) \in P_G\}.$$

For each $i \in V$, define its descendant set by

$$\mathrm{Desc}_G(i) = \{j \in V : (i, j) \in P_G\}.$$

Definition 1 (Effective burden and effective reach)**.** The *effective supervisory burden* of position j in G is

$$a_G(j) = |\,\mathrm{Anc}_G(j)|.$$

The *effective subordinate reach* of position i in G is

$$r_G(i) = |\,\mathrm{Desc}_G(i)|.$$

The corresponding labeled profiles are

$$\mathbf{a}(G) = (a_G(i))_{i \in V} \quad \text{and} \quad \mathbf{r}(G) = (r_G(i))_{i \in V}.$$

Their decreasing rearrangements are denoted by

$$\mathbf{a}^{\downarrow}(G) = \big(a_{(1)}(G), \ldots, a_{(n)}(G)\big) \quad \text{and} \quad \mathbf{r}^{\downarrow}(G) = \big(r_{(1)}(G), \ldots, r_{(n)}(G)\big).$$

The burden and reach profiles are the two labeled margins of the effective authority relation. The burden profile records the number of direct and indirect supervisors of each position, while the reach profile records the number of direct and indirect subordinates of each position.

Both profiles count distinct superior–subordinate pairs rather than directed paths. Multiple chains of command connecting the same superior and subordinate contribute only one effective authority relation. Consequently, an additional reporting link changes the profiles only if it creates at least one new effective superior–subordinate pair.

Lemma 1 (Common-volume identity)**.** *For every $G \in \mathcal{H}(V)$,*

$$\sum_{j \in V} a_G(j) = |P_G| = \sum_{i \in V} r_G(i).$$

Proof. Every effective authority relation contributes one unit to the burden of exactly one subordinate position and one unit to the reach of exactly one supervisory position. Hence both sums equal $|P_G|$. ∎

Thus burden and reach do not measure different aggregate quantities. They are different distributions of the same volume of effective authority relations.

3 Two-Sided Cumulative Dominance

3.1 Absolute dominance

For $k \in \{1, \dots, n\}$, define the cumulative burden and reach functions

$$A_k(G) = \sum_{s=1}^{k} a_{(s)}(G) \quad \text{and} \quad R_k(G) = \sum_{s=1}^{k} r_{(s)}(G).$$

Thus $A_k(G)$ is the total effective supervisory burden borne by the k most subordinated positions, while $R_k(G)$ is the total effective subordinate reach exercised by the k most authoritative positions.

Definition 2 (Absolute two-sided cumulative dominance)**.** For $G, H \in \mathcal{H}(V)$, write $G \succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} H$ if

$$A_k(G) \geq A_k(H) \quad \text{and} \quad R_k(G) \geq R_k(H), \qquad k \in \{1, \dots, n\}.$$

Write $G \succ_{\mathrm{AR}}^{\mathrm{abs}} H$ if $G \succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} H$ and $H \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} G$.

Equivalently, $G \succ_{\mathrm{AR}}^{\mathrm{abs}} H$ if and only if all the defining inequalities hold and at least one is strict. The superscript "abs" indicates that burden and reach are compared in their original units, so equal numerical differences in effective authority relations receive equal weight.

Absolute dominance requires simultaneous weak majorization of the burden and reach profiles. At the terminal cutoff, the common-volume identity gives $A_n(G) = R_n(G) = |P_G|$. Hence the two terminal conditions both reduce to $|P_G| \geq |P_H|$. The smaller cutoffs impose distributional restrictions on burden and reach, as discussed in Section 4.

Proposition 1 (Order properties)**.** *Absolute two-sided cumulative dominance is a preorder on $\mathcal{H}(V)$. It is invariant under relabeling of positions and under changes in the edge set that leave the reachability relation unchanged.*

Proof. Reflexivity is immediate, and transitivity follows from transitivity of the cumulative inequalities. Relabeling does not change the decreasing burden and reach profiles, while hierarchies with the same reachability relation have identical profiles. The two invariance claims follow. ∎

The symmetric part of $\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}}$ identifies hierarchies with the same anonymous burden and reach profiles. Distinct reachability relations may therefore be equivalent: the preorder retains the two anonymous margins rather than the full relation.

3.2 Proportional dominance

Absolute dominance evaluates numerical differences in effective authority relations. The proportional benchmark instead evaluates inclusive counts

$$\widetilde{a}_G(i) = 1 + a_G(i) \quad \text{and} \quad \widetilde{r}_G(i) = 1 + r_G(i),$$

where $\widetilde{a}_G(i)$ counts position i together with all its direct and indirect supervisors and $\widetilde{r}_G(i)$ counts position i together with all its direct and indirect subordinates.

For $k \in \{1, \dots, n\}$, define

$$A_k^{\log}(G) = \sum_{s=1}^{k} \log\big(1 + a_{(s)}(G)\big) \quad \text{and} \quad R_k^{\log}(G) = \sum_{s=1}^{k} \log\big(1 + r_{(s)}(G)\big).$$

Definition 3 (Proportional two-sided cumulative dominance)**.** For $G, H \in \mathcal{H}(V)$, write $G \succcurlyeq_{\mathrm{AR}}^{\mathrm{prop}} H$ if

$$A_k^{\log}(G) \geq A_k^{\log}(H) \quad \text{and} \quad R_k^{\log}(G) \geq R_k^{\log}(H), \qquad k \in \{1, \dots, n\}. \tag{1}$$

Write $G \succ_{\mathrm{AR}}^{\mathrm{prop}} H$ if $G \succcurlyeq_{\mathrm{AR}}^{\mathrm{prop}} H$ and $H \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{prop}} G$.

Equivalently, $G \succ_{\mathrm{AR}}^{\mathrm{prop}} H$ if and only if all the defining inequalities hold and at least one is strict. By monotonicity of the logarithm, (1) is equivalent to

$$\prod_{s=1}^{k} \big(1 + a_{(s)}(G)\big) \geq \prod_{s=1}^{k} \big(1 + a_{(s)}(H)\big)$$

and

$$\prod_{s=1}^{k} \big(1 + r_{(s)}(G)\big) \geq \prod_{s=1}^{k} \big(1 + r_{(s)}(H)\big)$$

for every $k \in \{1, \dots, n\}$.

The proportional criterion therefore compares geometric means of inclusive burden and reach within every upper-ranked group. It permits proportional advantages at some ranks to offset proportional disadvantages at others within the same cutoff, but it permits no cumulative proportional deficit at any cutoff.

Proposition 2 (Order properties)**.** *Proportional two-sided cumulative dominance is a preorder on $\mathcal{H}(V)$. It is invariant under relabeling of positions and under changes in the edge set that leave the reachability relation unchanged.*

Proof. Reflexivity is immediate, and transitivity follows from transitivity of the transformed cumulative inequalities. Relabeling does not change the ordered profiles, while hierarchies with the same reachability relation have identical transformed profiles. The two invariance claims follow. ∎

The symmetric part of $\succcurlyeq_{\mathrm{AR}}^{\mathrm{prop}}$ likewise identifies hierarchies with the same anonymous burden and reach profiles. The logarithmic transformation changes the comparison scale but not the information used: the relation remains insensitive to labels, redundant edges, and the association between burden and reach across positions.

4 Conceptual interpretation and robust unanimity

The defining principle can be stated without the language of majorization:

> An organizational hierarchy cannot be judged less hierarchical when, at every rank cutoff, it has at least as much effective subjection among its most subordinated positions and at least as much effective authority among its most authoritative positions.

This principle considers both the total number of effective authority relations and their distribution across positions. It ranks one hierarchy above another only when the burden and reach comparisons support the same ranking.

4.1 Volume and two-sided stratification

Every effective superior–subordinate pair is an instance of organizational authority. Because burden and reach count the opposite endpoints of the same pairs, their aggregate totals coincide. Under absolute dominance, $|P_G| > |P_H|$ means that G contains more effective authority relations than H.

Volume alone does not determine vertical structure. Equal numbers of effective authority relations can be distributed very differently across positions. The sequence

$$A_1(G), \ldots, A_n(G)$$

records the effective subjection accumulated by progressively larger groups of the most subordinated positions. Larger values at small cutoffs indicate that supervisory burden is concentrated among these positions. Similarly,

$$R_1(G), \ldots, R_n(G)$$

records the subordinate reach accumulated by progressively larger groups of the most authoritative positions. Larger values at small cutoffs indicate that authority is concentrated among these positions.

Neither perspective is sufficient on its own. Concentrated subjection may result from depth, convergent supervision, or other features of the reachability relation without being accompanied by concentrated authority. Conversely, concentrated authority may coexist with a more even distribution of subjection.

Two-sided dominance therefore requires agreement between the burden and reach comparisons. At every rank cutoff, a dominating hierarchy has no less cumulative burden among its most subordinated positions and no less cumulative reach among its most authoritative positions. When the two profiles support opposite rankings, or when the cumulative comparisons cross within either profile, the hierarchies remain incomparable. Resolving such incomparability requires an additional judgment about the relative importance of burden and reach or the tradeoffs permitted between them.

4.2 Absolute and proportional sensitivity

The absolute relation gives constant marginal significance to every effective authority relation. Raising reach from 1 to 2 and raising it from 100 to 101 each add one unit to the relevant cumulative sum. The proportional relation instead evaluates

$$\log(x+2) - \log(x+1) = \log\left(\frac{x+2}{x+1}\right),$$

which decreases with x. An additional effective relation therefore matters more when the position's inclusive domain is initially small.

Equivalently, equal proportional expansions of inclusive domains receive equal weight. Doubling $1 + x$ always raises its logarithm by $\log 2$. Thus the proportional benchmark treats equal percentage increases equally, whereas the absolute benchmark treats equal numerical increases equally.

The proportional benchmark is not merely the absolute benchmark expressed in different units. Concavity of the logarithm gives greater relative significance to counts at positions whose inclusive domains are smaller. Consider the profiles

$$\mathbf{x} = (3, 1, 1, 0) \quad \text{and} \quad \mathbf{y} = (2, 2, 1, 0).$$

Under the absolute scale, $\mathbf{x}$ weakly majorizes $\mathbf{y}$: its first coordinate is larger, while the two profiles have equal cumulative sums at every subsequent cutoff.

The proportional comparison is different. At the first cutoff, $\mathbf{x}$ remains ahead because $4 > 3$. At the second cutoff, however,

$$(1 + 3)(1 + 1) = 8 < 9 = (1 + 2)(1 + 2),$$

and at the terminal cutoff,

$$\prod_i (1 + x_i) = 16 < 18 = \prod_i (1 + y_i).$$

The proportional cumulative comparisons cross, so neither profile dominates the other on that scale. At the second cutoff, the more even distribution of the two largest counts in $\mathbf{y}$ produces the larger cumulative product, reversing the first-cutoff comparison. The two scales produce different comparisons: the absolute scale ranks $\mathbf{x}$ above $\mathbf{y}$, whereas the proportional scale leaves the profiles incomparable.

4.3 Robustness to numerical judgments

A numerical evaluation must determine how to value concentration within the burden and reach profiles, how to weight one profile against the other, and whether the association between burden and reach across positions matters. The product preorders compare concentration within each anonymous profile but remain neutral about the association between the two attributes. Hierarchies with the same burden and reach profiles are therefore treated as equivalent even if high burden and high reach occur at the same positions in one hierarchy and at different positions in the other.

A possible refinement would compare this association and treat a stronger separation between positions that exercise authority and positions that are subject to it as an additional form of hierarchical role polarization. Under that interpretation, hierarchy is more pronounced when high reach tends to occur at positions with low burden and high burden tends to occur at positions with low reach. The present relations do not impose this interpretation. Positions with both positive burden and positive reach may instead be viewed as intermediaries, coordinators, or organizational bottlenecks. Ranking hierarchies by the association between burden and reach therefore requires an additional organizational judgment.

The numerical interpretation uses separable marginal evaluations that are nondecreasing and convex in burden and reach measured on the relevant scale. Monotonicity treats additional effective authority or subjection as hierarchy enhancing. Convexity makes the evaluation responsive to concentration: holding the total on the relevant scale fixed, shifting burden or reach toward positions that already have higher values weakly raises the evaluation. Allowing every positive pair of burden and reach weights leaves the relative importance of the two profiles unrestricted.

The representation is established in Theorem 1. For either $s \in \{\text{abs}, \text{prop}\}$, $G \succcurlyeq^{s}_{\text{AR}} H$ if and only if every admissible separable marginal evaluation ranks G weakly above H. Hence, if G and H are incomparable, neither ranking is robust to the full admissible family. The product preorder leaves this disagreement unresolved rather than selecting a particular evaluation or priority rule. A nonseparable evaluation may rank additional pairs, but only by introducing a judgment about burden–reach association.

The representation applies to both profile scales. For absolute dominance, Theorem 2 establishes a stronger all-size result. Consider a population-independent index formed by assigning each position a score based on its burden and reach and summing these scores across positions. If the same local evaluation preserves every absolute product comparison at every finite population size, the resulting index admits an equivalent representation that is additively separable in burden and reach, with nondecreasing, discretely convex components. Thus, within this all-size position-additive class, every compatible index has a representation that is neutral about burden–reach association. No analogous exhaustiveness claim is made for proportional dominance.

5 Why both sides matter: a tree and a diamond

Figure 1 compares two hierarchies on four positions.

The ordered burden and reach profiles of the tree are

$$\mathbf{a}^{\downarrow}(H) = (2, 2, 1, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow}(H) = (3, 2, 0, 0),$$

while those of the diamond are

$$\mathbf{a}^{\downarrow}(G) = (3, 1, 1, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow}(G) = (3, 1, 1, 0).$$

Both hierarchies contain five effective authority relations. Thus, their difference under the absolute benchmark is distributional rather than volumetric.

The cumulative burden profiles are

$$\big(A_k(G)\big)_{k=1}^{4} = (3, 4, 5, 5) \quad \text{and} \quad \big(A_k(H)\big)_{k=1}^{4} = (2, 4, 5, 5).$$

The diamond weakly dominates the tree at every burden cutoff, with strict inequality at the first cutoff. Its bottom position w is subordinated to every other position, concentrating effective subjection in one position.

The cumulative reach profiles are

$$\big(R_k(G)\big)_{k=1}^{4} = (3, 4, 5, 5) \quad \text{and} \quad \big(R_k(H)\big)_{k=1}^{4} = (3, 5, 5, 5).$$

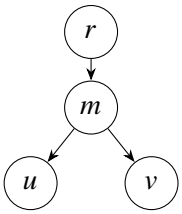
The tree weakly dominates the diamond at every reach cutoff, with strict inequality at the second cutoff. The intermediate manager m reaches two bottom positions, whereas the corresponding authority in the diamond is divided between p and q.

Therefore, the absolute one-sided criteria give opposite verdicts. The diamond concentrates subjection but disperses intermediate authority, while the tree concentrates intermediate authority but spreads deep subjection across two positions. Consequently, absolute two-sided cumulative dominance leaves the pair incomparable.

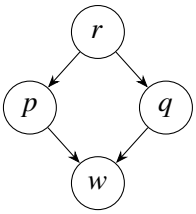
The proportional relation also leaves the pair incomparable, although for a different reason. On burden, the diamond has the larger leading inclusive count, since $4 > 3$, but the tree has the larger cumulative product at the second cutoff, since $9 > 8$. Therefore, the proportional burden comparisons cross. On reach, the hierarchies tie at the first cutoff, while the tree has the larger cumulative product at the second, since $12 > 8$. Hence neither hierarchy dominates the other.

This incomparability is consistent with the numerical representation discussed in Section 4.3 and established in Theorem 1. Neither strict ranking is unanimously supported by all admissible marginal evaluations and all positive burden–reach weights. For the absolute scale, the disagreement is visible using the separable quadratic evaluation

$$\alpha \sum_{v \in V} a_G(v)^2 + \beta \sum_{v \in V} r_G(v)^2, \qquad \alpha, \beta > 0.$$



Hierarchy H
Tree



Hierarchy G
Diamond

Figure 1: A tree and a diamond hierarchy.

The sums of squared burdens are 11 for G and 9 for H, while the sums of squared reaches are 11 for G and 13 for H. Hence the value for G minus the value for H is $2(\alpha - \beta)$. Thus G ranks above H when $\alpha > \beta$, H ranks above G when $\beta > \alpha$, and the two tie when $\alpha = \beta$. Selecting a strict ranking requires an organizational basis for assigning relative importance to burden and reach.

6 Basic properties

6.1 Chains and stars

Figure 2 compares a directed chain and an out-star on four positions.

The directed chain has

$$\mathbf{a}^{\downarrow} = (3, 2, 1, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow} = (3, 2, 1, 0),$$

while the out-star has

$$\mathbf{a}^{\downarrow} = (1, 1, 1, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow} = (3, 0, 0, 0).$$

The chain two-sidedly dominates the star. It contains more effective authority relations, imposes greater cumulative burden on its most subordinated positions, and sustains greater cumulative reach beyond the top position.

More generally, a directed chain on n positions has burden and reach profiles $(n-1, n-2, \ldots, 1, 0)$. Every pair of positions is comparable in its reachability relation, making the chain a natural upper benchmark. The precise extremal result is given below.

6.2 Incompleteness

Two-sided cumulative dominance can fail because the cumulative burden profiles cross, because the cumulative reach profiles cross, or because the two marginal comparisons support opposite rankings. The third source is distinctive to the product relation and records a conflict between concentration of subjection and concentration of authority.

6.3 Extremal hierarchies

Let E_n denote the empty hierarchy on n positions, and let C_n denote a directed chain on n positions.

Proposition 3 (Extremal hierarchies)**.** *On $\mathcal{H}(V)$, the empty hierarchy E_n is the unique minimum under $\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}}$. A hierarchy G is maximal under $\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}}$ if and only if its reachability relation P_G is a total order. Consequently, every hierarchy whose reachability relation is not total is strictly dominated by C_n under absolute dominance.*



Figure 2: A chain and an out-star on four positions.

Proof. The empty hierarchy has

$$\mathbf{a}^{\downarrow}(E_n) = \mathbf{r}^{\downarrow}(E_n) = (0, \dots, 0).$$

Nonnegativity of every burden and reach coordinate gives $G \succcurlyeq^{\text{abs}}_{\text{AR}} E_n$ for every $G \in \mathcal{H}(V)$. If $G \neq E_n$, then $|P_G| > 0$. By Lemma 1, both terminal cumulative inequalities are strict, so $G \succ^{\text{abs}}_{\text{AR}} E_n$.

Now choose a topological ordering $v_1, \dots, v_n$ of an arbitrary $G \in \mathcal{H}(V)$, meaning that every ancestor of a position precedes it in the ordering. Every ancestor of v_j belongs to $\{v_1, \dots, v_{j-1}\}$, so $a_G(v_j) \leq j - 1$. These coordinatewise bounds imply $a_{(k)}(G) \leq n - k$ for every $k \in \{1, \dots, n\}$. Applying the same argument to the reversed hierarchy gives $r_{(k)}(G) \leq n - k$. Since

$$\mathbf{a}^{\downarrow}(C_n) = \mathbf{r}^{\downarrow}(C_n) = (n-1, n-2, \dots, 1, 0),$$

we obtain $C_n \succcurlyeq^{\text{abs}}_{\text{AR}} G$.

If P_G is not total, some pair of positions is incomparable, and therefore $|P_G| < n(n-1)/2 = |P_{C_n}|$. Both terminal cumulative inequalities are then strict, so $C_n \succ^{\text{abs}}_{\text{AR}} G$. Conversely, if P_G is total, the ordered burden and reach profiles of G coincide with those of C_n, giving $G \sim^{\text{abs}}_{\text{AR}} C_n$. ∎

The same extremal hierarchies arise under proportional dominance. The coordinatewise bounds in Proposition 3 remain valid after the increasing transformation $x \mapsto \log(1 + x)$. Thus E_n is also the unique proportional minimum, and total reachability orders form the proportional maximal equivalence class.

A maximal DAG may contain redundant transitive edges in addition to the edges of a directed chain. Maximality requires only that every pair of distinct positions be comparable in the reachability relation.

6.4 Subordination Removal

Subordination Removal is inherited from the earlier operational framework and is used here as an external monotonicity test for the product preorders (Carbonell-Nicolau, 2025, 2026b). We recall the operation before establishing that both product relations satisfy it.

For each $j \in V$, let

$$\text{Sup}_G(j) = \{p \in V : (p, j) \in E_G\}$$

denote the set of immediate supervisors of j. For each edge $(j, i) \in E_G$, define

$$Q_G(j, i) = \left\{ p \in \text{Sup}_G(j) : \begin{array}{l} \text{there is no directed path from } p \text{ to } i \text{ in } G \\ \text{that avoids } j \end{array} \right\}.$$

Definition 4 (Subordination removal)**.** Let $G = (V, E_G) \in \mathcal{H}(V)$ and let $(j, i) \in E_G$. Removing the subordination relation (j, i) produces $G' = (V, E_{G'})$, where

$$E_{G'} = \big(E_G \setminus \{(j, i)\}\big) \cup \{(p, i) : p \in Q_G(j, i)\}.$$

The edge from j to i is deleted, and each immediate supervisor p of j is connected directly to i unless p already reaches i along a path avoiding j. If j has no immediate supervisor, no bypass edge is added. The resulting graph is acyclic (Carbonell-Nicolau, 2026b).

Figure 3 illustrates the operation. Removing $b \to c$ adds the bypass edge $a \to c$, which preserves the remaining authority relation from a to c.

An edge $(j, i) \in E_G$ is *redundant* if G contains a directed path from j to i other than the edge itself.

Definition 5 (Subordination Removal)**.** A dominance relation $\succcurlyeq$ satisfies *Subordination Removal* if every removal gives $G \succcurlyeq G'$. If the removed edge is nonredundant in G, the comparison must be strict: $G \succ G'$.

Proposition 4 (Consistency with Subordination Removal)**.** *Absolute and proportional two-sided cumulative dominance satisfy Subordination Removal.*

Proof. Suppose G' is obtained from G by removing (j, i). Every bypass edge (p, i) replaces the path $p \to j \to i$ already present in G. Replacing bypass edges in a path of G' by their corresponding two-edge paths implies that $P_{G'} \subseteq P_G$. Hence, for every $v \in V$,

$$a_{G'}(v) \le a_G(v) \quad \text{and} \quad r_{G'}(v) \le r_G(v).$$

The sum of the k largest coordinates is the maximum sum over all subsets of k positions. Since every labeled burden and reach coordinate weakly falls,

$$A_k(G) \ge A_k(G') \quad \text{and} \quad R_k(G) \ge R_k(G'), \qquad k \in \{1, \dots, n\}.$$

Thus $G \succcurlyeq^{\text{abs}}_{\text{AR}} G'$.

If the removed edge is nonredundant, then $P_{G'} \subsetneq P_G$. The common-volume identity makes both terminal inequalities strict, giving $G \succ^{\text{abs}}_{\text{AR}} G'$.

Because $x \mapsto \log(1+x)$ is strictly increasing, the same coordinatewise argument applies to the transformed profiles and gives $G \succcurlyeq^{\text{prop}}_{\text{AR}} G'$. Under a nonredundant removal, at least one burden coordinate and one reach coordinate fall strictly. Therefore, both terminal transformed inequalities are strict, so $G \succ^{\text{prop}}_{\text{AR}} G'$. ∎

A redundant removal may leave the reachability relation unchanged, in which case

$$G \sim^{\text{abs}}_{\text{AR}} G' \quad \text{and} \quad G \sim^{\text{prop}}_{\text{AR}} G'.$$

On domains without redundant edges, including the multipath-free domain studied by Carbonell-Nicolau (2026b), every removal strictly lowers hierarchy under both product relations.

6.5 Relation to sequential-removal dominance

Sequential-removal dominance is the central comparison rule of the multiple-supervisor framework in Carbonell-Nicolau (2026b). Every weak and strict sequential-removal comparison is preserved by both product relations, but the converse fails.

Let $\succcurlyeq_H$ denote the fixed-size dominance relation generated by finite sequences of Subordination Removal operations and relabelings. Thus $G \succcurlyeq_H H$ if, after a possible relabeling, H can be obtained from G by a finite removal sequence. Write $G \succ_H H$ if the sequence contains at least one nonredundant removal.

Proposition 5 (Consistency and proper extension)**.** *On every fixed-size DAG domain:*

1. *$G \succcurlyeq_H H$ implies $G \succcurlyeq^{\text{abs}}_{\text{AR}} H$ and $G \succcurlyeq^{\text{prop}}_{\text{AR}} H$;*
2. *$G \succ_H H$ implies $G \succ^{\text{abs}}_{\text{AR}} H$ and $G \succ^{\text{prop}}_{\text{AR}} H$;*



Before removal **After removal of $b \to c$**

Figure 3: Subordination removal. The edge $b \to c$ is deleted, and the bypass edge $a \to c$ preserves the remaining authority relation from a to c.

3. *both weak containments are proper:*

$$\succcurlyeq_H \subsetneq \succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} \quad \textit{and} \quad \succcurlyeq_H \subsetneq \succcurlyeq_{\mathrm{AR}}^{\mathrm{prop}} .$$

Proof. If $G \succcurlyeq_H H$, then, after relabeling, there is a sequence $G = G^0, G^1, \ldots, G^m = H$ of subordination removals. Proposition 4, transitivity, and relabeling invariance imply both weak product comparisons. If the sequence contains a nonredundant removal, one step is strict under both product relations, giving both strict comparisons.

For properness, let C be the chain $1 \to 2 \to 3$, and let D have edges $1 \to 3$ and $2 \to 3$. Their ordered profiles are

$$\mathbf{a}^{\downarrow}(C) = \mathbf{r}^{\downarrow}(C) = (2, 1, 0),$$

and

$$\mathbf{a}^{\downarrow}(D) = (2, 0, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow}(D) = (1, 1, 0).$$

The cumulative burden and reach sums are $(2, 3, 3)$ for C, compared with $(2, 2, 2)$ and $(1, 2, 2)$ for D. Hence $C \succ_{\mathrm{AR}}^{\mathrm{abs}} D$. The corresponding cumulative products of inclusive burden and reach are $(3, 6, 6)$ for C, compared with $(3, 3, 3)$ and $(2, 4, 4)$ for D. Hence $C \succ_{\mathrm{AR}}^{\mathrm{prop}} D$.

However, D cannot be obtained from C by subordination removals. Removing $1 \to 2$ leaves only $2 \to 3$, while removing $2 \to 3$ adds the bypass $1 \to 3$ and leaves the out-star with edges $1 \to 2$ and $1 \to 3$. Further removals cannot create two incomparable immediate supervisors of position 3. Thus $C \not\succcurlyeq_H D$, proving both proper containments. ∎

Thus, sequential-removal dominance is a common proper subrelation of the two product preorders. The absolute and proportional relations need not themselves be nested, because the logarithmic transformation can reverse a raw-count concentration comparison, as Section 4 shows.

7 Cross-size comparisons and replication principles

The preceding analysis compares hierarchies with a common number of positions. For each $n \geq 1$, let $\mathcal{H}_n$ denote the set of hierarchies on a fixed canonical n-element position set. Relabeling invariance allows any hierarchy on an arbitrary n-element set to be identified with a representative in $\mathcal{H}_n$. Let

$$\mathcal{H} = \bigcup_{n \geq 1} \mathcal{H}_n$$

denote the domain of all finite hierarchies. Disjoint unions, replications, and attachments use position-disjoint labeled copies, but the resulting comparisons do not depend on their labels.

Variable-size comparison requires an interpretation of population. Rank cutoffs may refer to the same number of positions, the same population fraction, or positions evaluated relative to the number of potential hierarchical counterparts. These interpretations are distinct from the profile scale. Let

$$\mathcal{S} = \{\text{abs}, \text{prop}\},$$

with

$$u_{\text{abs}}(x) = x \quad \text{and} \quad u_{\text{prop}}(x) = \log(1 + x).$$

For $s \in \mathcal{S}$, write $\succcurlyeq^{s}_{\text{AR}}$ for the corresponding fixed-size relation.

7.1 Disjoint replication

The replication principles below adapt the index principles of Carbonell-Nicolau (2026a) to hierarchical preorders. For $G = (V, E) \in \mathcal{H}$ and an integer $q \geq 1$, let G^q denote the disjoint union of q labeled copies of G, with

$$V(G^q) = V \times \{1, \ldots, q\}$$

and

$$E(G^q) = \big\{((i, \ell), (j, \ell)) : (i, j) \in E,\ \ell \in \{1, \ldots, q\}\big\}.$$

Replication preserves every effective authority relation within each copy and creates none between copies. Therefore, the burden and reach profiles of G^q repeat each coordinate of the corresponding profile of G exactly q times. The same conclusion holds after either transformation u_s.

Definition 6 (Intensive Replication Principle)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies the *Intensive Replication Principle* if $G^q \sim G$ for every $G \in \mathcal{H}$ and every integer $q \geq 1$.

The intensive interpretation evaluates hierarchy per position. Consequently, an organization and any number of disconnected replicas are equivalent.

Definition 7 (Extensive Replication Principle)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies the *Extensive Replication Principle* if $G^q \succ G$ for every G with $E(G) \neq \varnothing$ and every integer $q > 1$.

The extensive interpretation evaluates aggregate hierarchy. Replication reproduces every effective authority relation in each copy and strictly raises aggregate hierarchy whenever the original hierarchy is nonflat.

Definition 8 (Concentration Replication Principle)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies the *Concentration Replication Principle* if $G \succ G^q$ for every G with $E(G) \neq \varnothing$ and every integer $q > 1$.

The concentration interpretation evaluates hierarchy relative to the number of potential interpersonal relations. Replication increases the number of potential cross-copy pairs without creating authority relations between copies, so concentration falls.

For every nonflat G and $q > 1$, the three principles respectively require

$$G^q \sim G, \qquad G^q \succ G, \qquad G \succ G^q.$$

Thus, they are alternative population interpretations rather than axioms to be imposed jointly.

7.2 Two dimensions of cross-size evaluation

The profile scale determines whether burden and reach are evaluated absolutely or proportionally. The population interpretation determines whether they are compared intensively, extensively, or by concentration. The resulting relations are denoted by

$$\succcurlyeq_{\mathrm{AR}}^{s,X}, \qquad s \in \{\mathrm{abs}, \mathrm{prop}\}, \quad X \in \{I, E, C\}.$$

The analysis is organized by population interpretation, with the two profile scales treated in parallel. For each interpretation, we examine preorder properties, fixed-size agreement, replication, and structural extension. For the intensive and extensive interpretations, we also study uniqueness.

7.3 Structural extension principles

The attachment operations below adapt those of Carbonell-Nicolau (2026a) to hierarchical preorders. Throughout, component hierarchies are position-disjoint. A *root* has no immediate supervisor, a *leaf* has no immediate subordinate, and depth is effective supervisory burden. A *cross-edge* joins positions in different components.

Every cross-edge is directed from the designated upper component to the designated lower component. A path entering the lower component cannot return to the upper component, so all attachment operations preserve acyclicity. Figure 4 illustrates upward, downward, and partial attachment in panels (a), (b), and (c).

7.3.1 Unrestricted upward extension

For position-disjoint $G^o, G \in \mathcal{H}$, let $G^o \oplus G$ be the set of hierarchies obtained from their disjoint union by adding edges from leaves of G^o to roots of G, such that every leaf of G^o has an outgoing cross-edge and no other cross-edges are added. Thus G^o is appended above the base G; not every root of G need receive an edge. The set $G^o \oplus G$ is nonempty: selecting any root of G and connecting every leaf of G^o to that root produces an admissible attachment. See Figure 4, panel (a).

Definition 9 (Upward Extension)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies *Upward Extension* if $G^* \succ G$ for all position-disjoint $G^o, G \in \mathcal{H}$ and every $G^* \in G^o \oplus G$.

7.3.2 Unrestricted downward extension

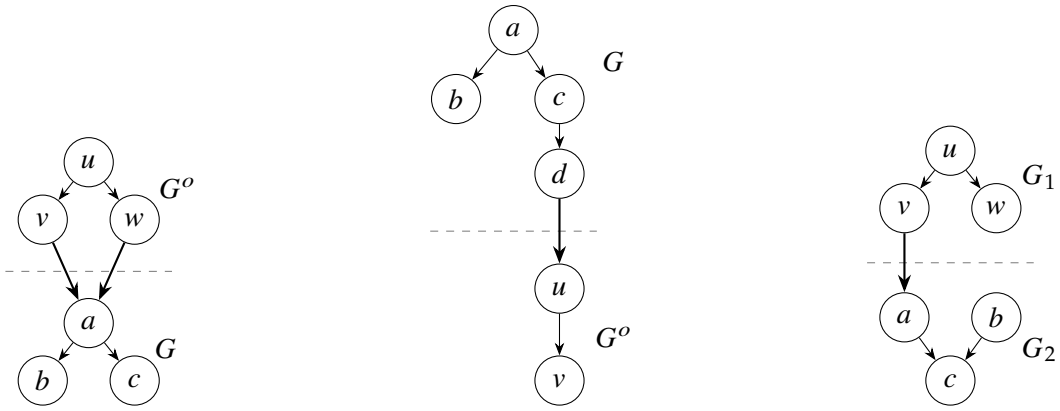
For position-disjoint $G, G^o \in \mathcal{H}$, let $G \boxplus G^o$ be the set of hierarchies obtained from their disjoint union by adding edges from maximum-depth leaves of G to roots of G^o, such that every root of G^o receives a cross-edge and no other cross-edges are added. Thus G^o is appended below the base G; not every maximum-depth leaf of G need have an outgoing edge.

A maximum-depth position is necessarily a leaf, since an immediate subordinate would have greater depth. Hence $G \boxplus G^o$ is nonempty: one may connect a maximum-depth leaf of G to every root of G^o. See Figure 4, panel (b).

Definition 10 (Downward Extension)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies *Downward Extension* if $G^* \succ G$ for all position-disjoint $G, G^o \in \mathcal{H}$ and every $G^* \in G \boxplus G^o$.

7.3.3 Conditional extensions

For position-disjoint $G_1, G_2 \in \mathcal{H}$, let $G_1 \uplus G_2$ be the set of hierarchies obtained from their disjoint union by adding a nonempty set of edges from leaves of G_1 to roots of G_2. The attachment may be partial: not every



(a) Upward attachment $G^* \in G^o \oplus G$

(b) Downward attachment $G^* \in G \boxplus G^o$

(c) Partial attachment $G^* \in G_1 \uplus G_2$

Figure 4: Structural attachments. Dashed lines separate component hierarchies, and thick arrows represent cross-edges. Panel (a) shows upward attachment, panel (b) downward attachment through a maximum-depth leaf of the base, and panel (c) partial attachment.

leaf of G_1 need send an edge, and not every root of G_2 need receive one. The set is nonempty because every finite nonempty DAG has a root and a leaf. See Figure 4, panel (c).

The unrestricted attachment sets satisfy

$$G^o \oplus G \subseteq G^o \uplus G \quad \text{and} \quad G \boxplus G^o \subseteq G \uplus G^o.$$

Definition 11 (Upward Extension*)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies *Upward Extension** if, for all position-disjoint $G^o, G \in \mathcal{H}$ and every $G^* \in G^o \uplus G$,

$$G^o \succcurlyeq G \Rightarrow G^* \succ G.$$

Definition 12 (Downward Extension*)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies *Downward Extension** if, for all position-disjoint $G, G^o \in \mathcal{H}$ and every $G^* \in G \uplus G^o$,

$$G^o \succcurlyeq G \Rightarrow G^* \succ G.$$

In both conditional principles, G is the base and G^o the appended hierarchy, while the premise is evaluated by the same preorder whose extension behavior is under study. Replication determines how population size enters the comparison; the extension principles test whether vertical attachment raises hierarchy under that interpretation.

8 The intensive extension

The intensive interpretation compares hierarchy per organizational position. Hierarchies of different sizes are replicated to a common population and then compared by the relevant fixed-size relation. At a common population N, each cutoff k represents the same population fraction k/N in both hierarchies.

Definition 13 (Intensive two-sided dominance)**.** Let $s \in \mathcal{S}$, let $G, H \in \mathcal{H}$, and write $n = |V(G)|$ and $m = |V(H)|$. Define

$$G \succcurlyeq^{s,I}_{\mathrm{AR}} H \Leftrightarrow G^m \succcurlyeq^{s}_{\mathrm{AR}} H^n.$$

The strict and symmetric parts are denoted by $\succ^{s,I}_{\mathrm{AR}}$ and $\sim^{s,I}_{\mathrm{AR}}$.

The replicated hierarchies have the common population nm. The next lemma shows that the comparison is unchanged by common replication.

Lemma 2 (Common replication)**.** *Let $G, H \in \mathcal{H}_n$, let $q \geq 1$, and let $s \in \mathcal{S}$. Then*

$$G \succcurlyeq^s_{\mathrm{AR}} H \Leftrightarrow G^q \succcurlyeq^s_{\mathrm{AR}} H^q.$$

The same equivalence holds for the strict and symmetric parts.

Proof. Consider either transformed profile, and let

$$x_1 \geq \cdots \geq x_n \quad \text{and} \quad y_1 \geq \cdots \geq y_n$$

be the ordered coordinates for G and H. Replication repeats each coordinate q times.

For a nonterminal cutoff $k < nq$, write $k = jq + r$, where $0 \leq j \leq n-1$ and $0 \leq r < q$. Using the convention that an empty sum equals zero, the cumulative sum of the repeated x-profile is

$$q \sum_{i=1}^{j} x_i + r x_{j+1} = (q - r) \sum_{i=1}^{j} x_i + r \sum_{i=1}^{j+1} x_i.$$

Thus the original cumulative inequalities at cutoffs j and $j+1$, with the cutoff-zero inequality understood as equality, imply every nonterminal inequality for the repeated profiles. At the terminal cutoff $k = nq$, the repeated sum is q times the original full sum.

Conversely, evaluating the repeated comparison at $q, 2q, \ldots, nq$ and dividing by q recovers the original cumulative inequalities. Applying the argument separately to burden and reach establishes the equivalence of the weak comparisons. Applying this equivalence in both directions shows that $G \sim^s_{\mathrm{AR}} H$ if and only if $G^q \sim^s_{\mathrm{AR}} H^q$. It also shows that $G \succ^s_{\mathrm{AR}} H$ if and only if $G^q \succ^s_{\mathrm{AR}} H^q$, since strict dominance is weak dominance without the reverse weak comparison. ∎

Proposition 6 (Properties and uniqueness of the intensive extension)**.** *For each $s \in \mathcal{S}$, the relation $\succcurlyeq^{s,I}_{\mathrm{AR}}$:*

1. *is a preorder on $\mathcal{H}$;*
2. *agrees with $\succcurlyeq^s_{\mathrm{AR}}$ on every $\mathcal{H}_n$;*
3. *is independent of the common multiple used to equalize population sizes;*
4. *satisfies the Intensive Replication Principle.*

Moreover, it is the unique preorder on $\mathcal{H}$ that agrees with $\succcurlyeq^s_{\mathrm{AR}}$ on every $\mathcal{H}_n$ and satisfies the Intensive Replication Principle.

Proof. Reflexivity is immediate. For transitivity, suppose $G \succcurlyeq^{s,I}_{\mathrm{AR}} H$ and $H \succcurlyeq^{s,I}_{\mathrm{AR}} K$, and write $|V(G)| = n$, $|V(H)| = m$, and $|V(K)| = \ell$. The premises give

$$G^m \succcurlyeq^s_{\mathrm{AR}} H^n \quad \text{and} \quad H^\ell \succcurlyeq^s_{\mathrm{AR}} K^m.$$

Replicating the first comparison ℓ times and the second n times, then applying fixed-size transitivity, gives $G^{m\ell} \succcurlyeq^s_{\mathrm{AR}} K^{mn}$. By Lemma 2, this is equivalent to $G^\ell \succcurlyeq^s_{\mathrm{AR}} K^n$, and hence $G \succcurlyeq^{s,I}_{\mathrm{AR}} K$.

If $G, H \in \mathcal{H}_n$, then

$$G \succcurlyeq^{s,I}_{\mathrm{AR}} H \Leftrightarrow G^n \succcurlyeq^s_{\mathrm{AR}} H^n \Leftrightarrow G \succcurlyeq^s_{\mathrm{AR}} H,$$

where the second equivalence follows from Lemma 2. This proves fixed-size agreement.

Let N be any common multiple of $n = |V(G)|$ and $m = |V(H)|$, and let $L = \operatorname{lcm}(N, nm)$. By common replication, comparison on the grid of length nm and comparison on the grid of length N are each equivalent to comparison on the common refinement of length L. Therefore,

$$G \succcurlyeq^{s,I}_{\mathrm{AR}} H \Leftrightarrow G^{N/n} \succcurlyeq^{s}_{\mathrm{AR}} H^{N/m}.$$

The definition is thus independent of the common multiple.

At the common population $q|V(G)|$, both G and G^q are represented by G^q. Hence $G^q \sim^{s,I}_{\mathrm{AR}} G$, proving Intensive Replication.

Finally, let $\succcurlyeq$ be any preorder satisfying fixed-size agreement and Intensive Replication. If $|V(G)| = n$ and $|V(H)| = m$, then $G \sim G^m$ and $H \sim H^n$. Consequently,

$$G \succcurlyeq H \Leftrightarrow G^m \succcurlyeq H^n \Leftrightarrow G^m \succcurlyeq^{s}_{\mathrm{AR}} H^n \Leftrightarrow G \succcurlyeq^{s,I}_{\mathrm{AR}} H.$$

Thus $\succcurlyeq = \succcurlyeq^{s,I}_{\mathrm{AR}}$. ∎

8.1 Structural extension under the intensive interpretation

The conditional extension principles follow from a property of disjoint unions. If one hierarchy intensively dominates another, adjoining the two as separate components produces a hierarchy that still weakly dominates the initially dominated hierarchy.

Lemma 3. *Let $G^o, G \in \mathcal{H}$ and $s \in \mathcal{S}$. If $G^o \succcurlyeq^{s,I}_{\mathrm{AR}} G$, then $(G^o, G) \succcurlyeq^{s,I}_{\mathrm{AR}} G$.*

Proof. Write $|V(G^o)| = p$ and $|V(G)| = q$. The premise gives $(G^o)^q \succcurlyeq^{s}_{\mathrm{AR}} G^p$. Weak majorization is preserved when the same vector is appended to both profiles.[1] Applying this observation to the transformed burden and reach profiles and adjoining the profiles of G^q gives

$$\left((G^o)^q, G^q\right) \succcurlyeq^{s}_{\mathrm{AR}} \left(G^p, G^q\right).$$

Up to relabeling,

$$\left((G^o)^q, G^q\right) \cong (G^o, G)^q \quad \text{and} \quad \left(G^p, G^q\right) \cong G^{p+q}.$$

Relabeling invariance gives

$$(G^o, G)^q \succcurlyeq^{s}_{\mathrm{AR}} G^{p+q}.$$

Since $|(G^o, G)| = p + q$ and $|V(G)| = q$, this is exactly the fixed-size comparison required by Definition 13 for $(G^o, G) \succcurlyeq^{s,I}_{\mathrm{AR}} G$. ∎

Proposition 7 (Conditional structural extensions)**.** *For each $s \in \mathcal{S}$, the intensive relation $\succcurlyeq^{s,I}_{\mathrm{AR}}$ satisfies Upward Extension* and Downward Extension*.*

[1] To see this, let X_ℓ and Z_ℓ denote the sums of the ℓ largest coordinates of vectors $\mathbf{x}$ and $\mathbf{z}$, respectively. The sum of the k largest coordinates of their concatenation is

$$\max_{\ell} \{X_\ell + Z_{k-\ell}\},$$

where the maximum is taken over the feasible values of ℓ. If $\mathbf{x}$ weakly majorizes $\mathbf{y}$, then $X_\ell \geq Y_\ell$ for every ℓ, and therefore

$$\max_{\ell} \{X_\ell + Z_{k-\ell}\} \geq \max_{\ell} \{Y_\ell + Z_{k-\ell}\}.$$

Thus $(\mathbf{x}, \mathbf{z})$ weakly majorizes $(\mathbf{y}, \mathbf{z})$.

Proof. Suppose $G^o \succcurlyeq^{s,I}_{\mathrm{AR}} G$. By Lemma 3, $(G^o, G) \succcurlyeq^{s,I}_{\mathrm{AR}} G$.

Let G^* be obtained by adding an admissible nonempty set of cross-edges to the disjoint union (G^o, G). These edges remove no effective authority relation and create at least one. Therefore, every labeled burden and reach coordinate weakly increases, with at least one strict increase in each profile. Since u_s is strictly increasing, $G^* \succ^{s}_{\mathrm{AR}} (G^o, G)$. The two hierarchies have the same population, so fixed-size agreement gives $G^* \succ^{s,I}_{\mathrm{AR}} (G^o, G)$. Transitivity with $(G^o, G) \succcurlyeq^{s,I}_{\mathrm{AR}} G$ yields $G^* \succ^{s,I}_{\mathrm{AR}} G$. This proves Upward Extension* when G^o is attached above G and Downward Extension* when G^o is attached below G. ∎

The unrestricted principles fail because attachment may increase effective authority while diluting the upper tail of the intensive profile.

Proposition 8 (Failure of unrestricted upward extension)**.** *Neither* $\succcurlyeq^{\mathrm{abs},I}_{\mathrm{AR}}$ *nor* $\succcurlyeq^{\mathrm{prop},I}_{\mathrm{AR}}$ *satisfies Upward Extension.*

Proof. Let the lower base G consist of the chain

$$1 \to 2 \to 3 \to 4$$

and an isolated position 5. Let G^o consist of one position u, and form $G^* \in G^o \oplus G$ by adding $u \to 5$. The ordered burden and reach profiles change from

$$(3, 2, 1, 0, 0) \quad \text{to} \quad (3, 2, 1, 1, 0, 0).$$

Under the absolute scale, the total burden and reach are both 6 in G and 7 in G^*. Intensive dominance compares $(G^*)^5$ with G^6. At the terminal cutoff, the required inequality is therefore

$$5 \cdot 7 \geq 6 \cdot 6,$$

or equivalently, $\frac{7}{6} \geq \frac{6}{5}$. Since this inequality fails, $G^* \not\succcurlyeq^{\mathrm{abs},I}_{\mathrm{AR}} G$.

For proportional dominance, compare $(G^*)^5$ with G^6 at cutoff $k = 6$. The transformed cumulative burden and reach values are $5\log 4 + \log 3$ for $(G^*)^5$ and $6\log 4$ for G^6. Since $\log 3 < \log 4$,

$$5\log 4 + \log 3 < 6\log 4.$$

Thus $G^* \not\succcurlyeq^{\mathrm{prop},I}_{\mathrm{AR}} G$. ∎

Proposition 9 (Failure of unrestricted downward extension)**.** *Neither* $\succcurlyeq^{\mathrm{abs},I}_{\mathrm{AR}}$ *nor* $\succcurlyeq^{\mathrm{prop},I}_{\mathrm{AR}}$ *satisfies Downward Extension.*

Proof. Let the nine-position base hierarchy G have edge set

$$\{1 \to 4,\ 1 \to 7,\ 2 \to 4,\ 2 \to 5,\ 2 \to 9,\ 3 \to 8,\ 6 \to 8\}.$$

Its ordered profiles are

$$\mathbf{a}^{\downarrow}(G) = (2, 2, 1, 1, 1, 0, 0, 0, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow}(G) = (3, 2, 1, 1, 0, 0, 0, 0, 0).$$

Positions 4 and 8 are maximum-depth leaves. Append a single lower position 10 by adding $8 \to 10$. The resulting hierarchy $G^* \in G \boxplus G^o$ has profiles

$$\mathbf{a}^{\downarrow}(G^*) = (3, 2, 2, 1, 1, 1, 0, 0, 0, 0) \quad \text{and} \quad \mathbf{r}^{\downarrow}(G^*) = (3, 2, 2, 2, 1, 0, 0, 0, 0, 0).$$

Compare $(G^*)^9$ with G^{10} at reach cutoff $k = 10$. The absolute cumulative values are

$$R_{10}\big((G^*)^9\big) = 29 \quad \text{and} \quad R_{10}(G^{10}) = 30.$$

Hence $G^* \not\succcurlyeq^{\mathrm{abs},I}_{\mathrm{AR}} G$. The corresponding transformed values are $9\log 4 + \log 3$ and $10\log 4$. Since $\log 3 < \log 4$,

$$9\log 4 + \log 3 < 10\log 4,$$

and therefore $G^* \not\succcurlyeq^{\mathrm{prop},I}_{\mathrm{AR}} G$. ■

In the downward-attachment example, average burden and reach rise from $7/9$ to 1, but intensive dominance nevertheless fails. The attachment leaves the maximum reach equal to 3 while reducing the population share of positions attaining that value from $1/9$ to $1/10$. After replication to a common population, the base has the larger cumulative reach at the relevant upper-tail cutoff. Intensive dominance compares the full cumulative profiles, not only their averages.

Thus, both intensive relations satisfy Upward Extension* and Downward Extension* but fail Upward Extension and Downward Extension.

9 The extensive extension

The extensive interpretation compares aggregate hierarchy. Rather than equalizing population shares by replication, it compares equal absolute numbers of positions. The smaller hierarchy is embedded in the larger population by adjoining isolated positions, whose burden and reach are zero under both profile scales.

For $G \in \mathcal{H}$ and an integer $N \geq |V(G)|$, let G^{+N} denote any hierarchy obtained by adjoining $N - |V(G)|$ isolated positions to G. The labels assigned to the new positions are immaterial.

Definition 14 (Extensive two-sided dominance)**.** Let $G, H \in \mathcal{H}$, let

$$N = \max\{|V(G)|, |V(H)|\},$$

and let $s \in \mathcal{S}$. Define $G \succcurlyeq^{s,E}_{\mathrm{AR}} H$ if $G^{+N} \succcurlyeq^{s}_{\mathrm{AR}} H^{+N}$. The strict and symmetric parts are denoted by $\succ^{s,E}_{\mathrm{AR}}$ and $\sim^{s,E}_{\mathrm{AR}}$.

Adjoining an isolated position appends one zero to both profiles. Because $u_{\mathrm{abs}}(0) = u_{\mathrm{prop}}(0) = 0$, the construction applies under either profile scale. Equivalently, both hierarchies are zero-extended to a common length and compared at every absolute rank cutoff.

Definition 15 (Isolated-Position Neutrality)**.** A preorder $\succcurlyeq$ on $\mathcal{H}$ satisfies *Isolated-Position Neutrality* if $G' \sim G$ whenever G' is obtained from G by adjoining finitely many isolated positions.

Proposition 10 (Properties and uniqueness of the extensive extension)**.** *For each $s \in \mathcal{S}$, the relation $\succcurlyeq^{s,E}_{\mathrm{AR}}$:*

1. *is a preorder on $\mathcal{H}$;*
2. *agrees with $\succcurlyeq^{s}_{\mathrm{AR}}$ on every fixed-size subdomain;*
3. *satisfies Isolated-Position Neutrality;*
4. *satisfies the Extensive Replication Principle.*

Moreover, it is the unique preorder on $\mathcal{H}$ that agrees with $\succcurlyeq^{s}_{\mathrm{AR}}$ on every fixed-size subdomain and satisfies Isolated-Position Neutrality.

Proof. Reflexivity is immediate. For transitivity, suppose $G \succcurlyeq^{s,E}_{\mathrm{AR}} H$ and $H \succcurlyeq^{s,E}_{\mathrm{AR}} K$, and let

$$N = \max\{|V(G)|, |V(H)|, |V(K)|\}.$$

Further zero-extension preserves every cumulative comparison. Hence

$$G^{+N} \succcurlyeq^{s}_{\mathrm{AR}} H^{+N} \quad \text{and} \quad H^{+N} \succcurlyeq^{s}_{\mathrm{AR}} K^{+N}.$$

Fixed-size transitivity gives $G^{+N} \succcurlyeq^{s}_{\mathrm{AR}} K^{+N}$ and therefore $G \succcurlyeq^{s,E}_{\mathrm{AR}} K$.

If G and H have the same size, no isolated positions are added, so $G \succcurlyeq^{s,E}_{\mathrm{AR}} H$ if and only if $G \succcurlyeq^{s}_{\mathrm{AR}} H$. If G' is obtained from G by adjoining isolated positions, zero-extending G to the population of G' produces a relabeling of G'. Hence $G' \sim^{s,E}_{\mathrm{AR}} G$.

Now let G be nonflat and let $q > 1$. At population $q|V(G)|$, the zero-extension of G has one copy of each transformed coordinate of G followed by zeros, while G^q has q copies of each coordinate. Every cumulative sum for the replicated profile is therefore weakly greater than the corresponding sum for the zero-extended profile. Because both the burden and reach profiles of G contain a positive coordinate, both terminal transformed totals are strictly greater for G^q. Thus $G^q \succ^{s,E}_{\mathrm{AR}} G$, proving Extensive Replication.

Finally, let $\succcurlyeq$ be any preorder satisfying fixed-size agreement and Isolated-Position Neutrality. For arbitrary G, H, let $N = \max\{|V(G)|, |V(H)|\}$. Then $G \sim G^{+N}$ and $H \sim H^{+N}$, so

$$G \succcurlyeq H \Leftrightarrow G^{+N} \succcurlyeq H^{+N} \Leftrightarrow G^{+N} \succcurlyeq^{s}_{\mathrm{AR}} H^{+N} \Leftrightarrow G \succcurlyeq^{s,E}_{\mathrm{AR}} H.$$

Hence $\succcurlyeq = \succcurlyeq^{s,E}_{\mathrm{AR}}$. ∎

9.1 Structural extension under the extensive interpretation

Under extensive dominance, an appended component contributes nonnegative burden and reach coordinates, while every cross-edge creates additional effective authority relations. Consequently, every admissible vertical attachment strictly raises hierarchy relative to the base.

Proposition 11 (Structural extensions under extensive dominance)**.** *For each $s \in \mathcal{S}$, the extensive relation $\succcurlyeq^{s,E}_{\mathrm{AR}}$ satisfies Upward Extension, Downward Extension, Upward Extension*, and Downward Extension*.*

Proof. Let G be the base hierarchy, let G^o be a position-disjoint appended hierarchy, and let G^* be obtained from their disjoint union by any admissible nonempty set of cross-edges directed from the designated upper component to the designated lower component. The argument applies to $\oplus$, $\boxplus$, and $\uplus$.

Let $\overline{G}$ be obtained from G by adjoining $|V(G^o)|$ isolated positions. Then $\overline{G}$ and G^* have the same population, and Isolated-Position Neutrality gives $\overline{G} \sim^{s,E}_{\mathrm{AR}} G$.

Match each position of G in $\overline{G}$ with the corresponding position of G in G^*, and match the isolated positions in $\overline{G}$ with the positions of G^o. For the first group, adding G^o and the cross-edges removes no effective authority relation, so the burden and reach of each position of G weakly increase. For the second group, the matched coordinates in $\overline{G}$ are both zero, whereas burden and reach in G^* are nonnegative. Hence every burden and reach coordinate of G^* is weakly greater than its matched coordinate in $\overline{G}$.

Because the nonempty set of cross-edges creates at least one new effective authority relation, at least one position gains burden and at least one position gains reach. Thus each profile has at least one strict increase. Since u_s is strictly increasing, $G^* \succ^{s}_{\mathrm{AR}} \overline{G}$. The two hierarchies have the same population, so fixed-size agreement gives $G^* \succ^{s,E}_{\mathrm{AR}} \overline{G}$. Transitivity with $\overline{G} \sim^{s,E}_{\mathrm{AR}} G$ yields $G^* \succ^{s,E}_{\mathrm{AR}} G$.

This proves Upward Extension and Downward Extension. It also proves Upward Extension* and Downward Extension* without using their dominance premises. ∎

Thus the extensive relations satisfy all four structural extension principles. Unlike the intensive interpretation, aggregate hierarchy is not diluted by population growth: the appended component contributes nonnegative coordinates, and the cross-edges make both terminal profile comparisons strict.

10 The concentration extension

The concentration interpretation evaluates burden and reach relative to what is possible at the relevant population size. Cross-size comparison requires two adjustments. Repetition places the profiles on a common population grid, so each cutoff represents the same fraction of positions. Normalization divides each transformed burden and reach coordinate by its largest feasible value at that population size, so that every normalized coordinate lies in $[0, 1]$.

This analysis differs in status from the intensive and extensive analyses. Fixed-size agreement and Concentration Replication do not single out a unique cross-size relation. The construction below selects one natural relation by normalizing each position's burden and reach before comparing the profiles. We then study its replication, structural-extension, and numerical-representation properties.

For a hierarchy G with $n \geq 2$ positions and $s \in \mathcal{S}$, define

$$\widehat{a}^s_G(i) = \frac{u_s(a_G(i))}{u_s(n-1)} \quad \text{and} \quad \widehat{r}^s_G(i) = \frac{u_s(r_G(i))}{u_s(n-1)}.$$

For a singleton hierarchy, define both normalized profiles to be zero. Under the two profile scales,

$$\widehat{a}^{\text{abs}}_G(i) = \frac{a_G(i)}{n-1} \quad \text{and} \quad \widehat{r}^{\text{abs}}_G(i) = \frac{r_G(i)}{n-1},$$

while

$$\widehat{a}^{\text{prop}}_G(i) = \frac{\log(1 + a_G(i))}{\log n} \quad \text{and} \quad \widehat{r}^{\text{prop}}_G(i) = \frac{\log(1 + r_G(i))}{\log n}.$$

Every normalized coordinate belongs to $[0, 1]$. Under the absolute scale, it is the fraction of potential supervisors or subordinates represented in the position's effective domain. Under the proportional scale, it is the logarithm of the inclusive domain divided by the logarithm of the largest feasible inclusive domain.

Let

$$\widehat{\mathbf{a}}^{s,\downarrow}(G) = \big(\widehat{a}^s_{G,(1)}, \ldots, \widehat{a}^s_{G,(n)}\big) \quad \text{and} \quad \widehat{\mathbf{r}}^{s,\downarrow}(G) = \big(\widehat{r}^s_{G,(1)}, \ldots, \widehat{r}^s_{G,(n)}\big)$$

denote the decreasing normalized profiles. For a positive integer q, let $\widehat{\mathbf{a}}^{s,\downarrow}_q(G)$ and $\widehat{\mathbf{r}}^{s,\downarrow}_q(G)$ repeat each corresponding coordinate q times. Define

$$\widehat{A}^s_{q,k}(G) = \sum_{i=1}^{k} \widehat{a}^s_{G,q,(i)} \quad \text{and} \quad \widehat{R}^s_{q,k}(G) = \sum_{i=1}^{k} \widehat{r}^s_{G,q,(i)}.$$

Definition 16 (Concentration two-sided dominance)**.** Let $s \in \mathcal{S}$, let $G, H \in \mathcal{H}$, and write $n = |V(G)|$ and $m = |V(H)|$. Define $G \succcurlyeq^{s,C}_{\text{AR}} H$ if, for every $k \in \{1, \ldots, nm\}$,

$$\widehat{A}^s_{m,k}(G) \geq \widehat{A}^s_{n,k}(H) \quad \text{and} \quad \widehat{R}^s_{m,k}(G) \geq \widehat{R}^s_{n,k}(H).$$

The strict and symmetric parts are denoted by $\succ^{s,C}_{\text{AR}}$ and $\sim^{s,C}_{\text{AR}}$.

The repeated profiles have the common length nm, so cutoff k represents the same population fraction $k/(nm)$ in both hierarchies.

Proposition 12 (Properties of the concentration extension)**.** *For each $s \in \mathcal{S}$, the relation $\succcurlyeq^{s,C}_{\text{AR}}$:*

1. *is a preorder on $\mathcal{H}$;*
2. *agrees with $\succcurlyeq^{s}_{\mathrm{AR}}$ on every fixed-size subdomain;*
3. *is independent of the common multiple used to compare population fractions;*
4. *satisfies the Concentration Replication Principle.*

Proof. Reflexivity is immediate. To prove transitivity, suppose

$$G \succcurlyeq^{s,C}_{\mathrm{AR}} H \quad \text{and} \quad H \succcurlyeq^{s,C}_{\mathrm{AR}} K,$$

and write

$$|V(G)| = n, \qquad |V(H)| = m, \qquad |V(K)| = \ell.$$

The first premise compares the normalized profiles of G and H on a common grid of length nm, while the second compares those of H and K on a common grid of length $m\ell$.

Repeat every coordinate in the first comparison ℓ times and every coordinate in the second comparison n times. Both comparisons then lie on a common grid of length $nm\ell$, and the normalized profile of H is represented in the same way in both. Therefore, transitivity of the cumulative inequalities gives the burden and reach comparisons between G and K on that grid.

The resulting profiles are obtained by repeating each coordinate in the defining $n\ell$-grid comparison between G and K exactly m times. By Lemma 2, common repetition reflects as well as preserves cumulative comparisons. Removing the common repetition factor m gives the required burden and reach comparisons on the grid of length $n\ell$. Hence $G \succcurlyeq^{s,C}_{\mathrm{AR}} K$.

If G and H have the same size $n \geq 2$, their normalized transformed coordinates have the common positive denominator $u_s(n-1)$. Common positive rescaling does not affect cumulative comparisons, so

$$G \succcurlyeq^{s,C}_{\mathrm{AR}} H \Leftrightarrow G \succcurlyeq^{s}_{\mathrm{AR}} H.$$

The same conclusion holds for singleton hierarchies because all their profiles are zero.

Now let N be any common multiple of $n = |V(G)|$ and $m = |V(H)|$, and let $L = \operatorname{lcm}(N, nm)$. By common repetition, comparison on the grid of length nm and comparison on the grid of length N are both equivalent to comparison on the common refinement of length L. Therefore, the concentration comparison is independent of the common population grid.

Finally, let G have $n \geq 2$ positions and at least one edge, and let $q > 1$. Each position in every copy of G has the same raw burden and reach as the corresponding position in G, because replication creates no authority relations between copies. The population, however, increases from n to qn, so the normalization denominator increases from $u_s(n-1)$ to $u_s(qn-1)$. Hence, for every copied position i,

$$\frac{u_s(a_G(i))}{u_s(qn-1)} = \lambda_s(n,q)\frac{u_s(a_G(i))}{u_s(n-1)},$$

where

$$\lambda_s(n,q) = \frac{u_s(n-1)}{u_s(qn-1)} \in (0,1).$$

The same identity holds for reach.

To compare G and G^q, repeat the normalized profiles of G q times. Both profiles then have length qn. Relative to these repeated profiles of G, every normalized coordinate of G^q is multiplied by the common factor $\lambda_s(n,q) < 1$. Thus every normalized coordinate for G is weakly greater than the corresponding coordinate for G^q. Because G has at least one edge, each of its burden and reach profiles contains a positive coordinate, so the terminal cumulative inequality is strict for both profiles. Therefore, $G \succ^{s,C}_{\mathrm{AR}} G^q$. Thus Concentration Replication holds. ∎

10.1 Structural extension under the concentration interpretation

The concentration relations fail all four structural extension principles. For conditional attachment, a bounded number of new authority relations cannot offset the expansion of the population-specific normalization. For unrestricted attachment, the number of new relations grows with the appended population, but still too slowly relative to the normalization.

Proposition 13 (Failure of structural extensions under concentration)**.** *For each $s \in \mathcal{S}$, the concentration relation $\succcurlyeq_{\mathrm{AR}}^{s,C}$ satisfies none of Upward Extension, Downward Extension, Upward Extension*, or Downward Extension*.*

Proof. We first consider the conditional principles.

Failure of Upward Extension and Downward Extension*.* Fix a nonflat hierarchy B with $n \geq 2$ positions. For each $q \geq 1$, take position-disjoint copies U_q and L_q of B^q. Since they are isomorphic, $U_q \sim_{\mathrm{AR}}^{s,C} L_q$.

Select one copy of B in each hierarchy and add one cross-edge from a leaf i in the selected upper copy to a root j in the selected lower copy. Denote the resulting hierarchy by J_q, so $J_q \in U_q \uplus L_q$. The new effective authority relations run from i and its ancestors in the selected upper copy to j and its descendants in the selected lower copy. Because all other copies remain disconnected, the number of new effective authority relations depends only on B and the selected cross-edge, not on q. Denote this number by C.

Under the absolute scale, the total burden and total reach of B^q both equal $q|P_B|$. Since B^q has qn positions and normalization denominator $qn - 1$, its average normalized burden and reach are both

$$\frac{q|P_B|}{qn(qn-1)} = \frac{|P_B|}{n(qn-1)}.$$

The hierarchy J_q has $2qn$ positions and $2q|P_B| + C$ effective authority relations. Its average normalized burden and reach are therefore both

$$\frac{2q|P_B| + C}{2qn(2qn-1)}.$$

The ratio of this value to the corresponding value for either component is

$$\frac{(2q|P_B| + C)(qn-1)}{2q(2qn-1)|P_B|},$$

which converges to $1/2$ as $q \to \infty$. Hence, for all sufficiently large q, the average normalized burden and reach of J_q are strictly smaller than those of either component.

The hierarchy J_q has $2qn$ positions, while each component has qn positions. To place the normalized profiles on the common grid required by the definition, repeat every coordinate of the component profile twice. The preceding inequality implies that, for all sufficiently large q, the terminal normalized burden and reach sums of J_q are smaller than those of the twice-repeated component profiles. Hence J_q does not concentration-dominate either component under the absolute scale.

For the proportional scale, define

$$S_B^a = \sum_{i \in V(B)} \log\bigl(1 + a_B(i)\bigr).$$

Replication leaves the burden of every copied position unchanged. Therefore, each of the q copies contributes the transformed burden sum S_B^a. Since B^q has qn positions and normalization denominator $\log(qn)$, its average normalized burden is

$$\frac{qS_B^a}{qn\log(qn)} = \frac{S_B^a}{n\log(qn)}.$$

The same expression applies to either component U_q or L_q, since each is isomorphic to B^q.

When the cross-edge is added, only the root j and its descendants in the selected lower copy acquire additional supervisors. All positions in the other copies retain their original burdens. Let Δ_a denote the resulting increase in the transformed burden sum. Since the affected copy and the cross-edge are fixed, Δ_a depends on B and the selected cross-edge but not on q. The average normalized burden of J_q is therefore

$$\frac{2qS_B^a + \Delta_a}{2qn\log(2qn)}.$$

Because B is nonflat, $S_B^a > 0$. The inequality

$$\frac{2qS_B^a + \Delta_a}{2qn\log(2qn)} < \frac{S_B^a}{n\log(qn)}$$

is equivalent to

$$\frac{\Delta_a \log(qn)}{2q} < S_B^a \log 2.$$

The left side converges to zero as $q \to \infty$, so the inequality holds for all sufficiently large q.

To compare J_q, which has $2qn$ positions, with either component, which has qn positions, the definition repeats the component's normalized burden profile twice. Both profiles then have length $2qn$. At the terminal cutoff, their cumulative sums equal $2qn$ times their respective average normalized burdens. Since the average normalized burden of J_q is smaller for all sufficiently large q, the terminal burden inequality required for concentration dominance fails. Hence J_q does not concentration-dominate either component under the proportional scale.

The dominance premise of each conditional principle holds because $U_q \sim_{\mathrm{AR}}^{s,C} L_q$. Treating U_q as the upper appended hierarchy and L_q as the lower base yields a violation of Upward Extension*. Treating U_q as the upper base and L_q as the lower appended hierarchy yields a violation of Downward Extension*.

Failure of Upward Extension. Fix the same nonflat hierarchy B, with n positions, and let F_q be the flat hierarchy on q positions. Select a root b of B and add an edge from every position of F_q to b. Denote the resulting hierarchy by J_q^U, so $J_q^U \in F_q \oplus B$. Let

$$D_b = \{b\} \cup \mathrm{Desc}_B(b) \quad \text{and} \quad d = |D_b| = 1 + r_B(b).$$

Each of the q new upper positions reaches exactly the d positions in D_b. Consequently,

$$|P_{J_q^U}| = |P_B| + qd.$$

Under the absolute scale, the average normalized burden and reach of J_q^U are both

$$\frac{|P_B| + qd}{(n+q)(n+q-1)}.$$

This expression converges to zero as $q \to \infty$, whereas the corresponding value for B is

$$\frac{|P_B|}{n(n-1)} > 0.$$

Hence, for all sufficiently large q, $J_q^U \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs},C} B$.

For the proportional scale, define

$$S_B^r = \sum_{i \in V(B)} \log\bigl(1 + r_B(i)\bigr).$$

The transformed reach sum of J_q^U is

$$S_B^r + q\log(1+d),$$

because each new upper position has reach d, while the reach counts of positions in B remain unchanged. Its average normalized reach is therefore

$$\frac{S_B^r + q\log(1+d)}{(n+q)\log(n+q)},$$

which converges to zero as $q \to \infty$. Since B is nonflat, its average normalized reach is positive. The terminal reach inequality therefore fails for all sufficiently large q, so $J_q^U \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{prop},C} B$. Upward Extension fails under both profile scales.

Failure of Downward Extension. Again let F_q be flat on q positions. Select a maximum-depth leaf ℓ of B and add an edge from ℓ to every position of F_q. Denote the resulting hierarchy by J_q^D, so $J_q^D \in B \boxplus F_q$. Let

$$A_\ell = \{\ell\} \cup \mathrm{Anc}_B(\ell) \quad \text{and} \quad c = |A_\ell| = 1 + a_B(\ell).$$

Each new lower position has exactly the c supervisors in A_ℓ. Consequently,

$$|P_{J_q^D}| = |P_B| + cq.$$

Under the absolute scale, the average normalized burden and reach of J_q^D are both

$$\frac{|P_B| + cq}{(n+q)(n+q-1)},$$

which converges to zero. Since the corresponding value for B is positive, for all sufficiently large q, $J_q^D \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs},C} B$.

Under the proportional scale, each new lower position has burden c, while the burden counts of positions in B remain unchanged. Define

$$S_B^a = \sum_{i \in V(B)} \log\bigl(1 + a_B(i)\bigr).$$

The transformed burden sum of J_q^D is

$$S_B^a + q\log(1+c).$$

Therefore, its average normalized burden is

$$\frac{S_B^a + q\log(1+c)}{(n+q)\log(n+q)},$$

which converges to zero. Since B is nonflat, its average normalized burden is positive. The terminal burden inequality therefore fails for all sufficiently large q, so $J_q^D \not\succcurlyeq_{\mathrm{AR}}^{\mathrm{prop},C} B$. Downward Extension fails under both profile scales. ∎

Thus both concentration relations fail UE, DE, UE*, and DE*. In every case, the gains from attachment grow more slowly than the population-specific normalization.

11 Numerical representation and robust unanimity

The product preorders are incomplete because cumulative comparisons may cross within a profile or conflict across burden and reach. A single real-valued index must resolve such cases through additional judgments about concentration, burden–reach tradeoffs, or burden–reach association. The product preorders instead retain only comparisons supported unanimously by a family of marginal evaluations.

This section establishes the corresponding robust-unanimity representation and connects it with the depth–span framework of Carbonell-Nicolau (2026a). The earlier paper supplies the average, total, and concentration index architectures. The unanimity representations, Robust Marginal Unanimity, and the maximal absolute local-index theorem are specific to the product preorders introduced here.

11.1 Separable local evaluations

Let C denote the class of nondecreasing convex functions from $\mathbb{R}_+$ to $\mathbb{R}$. For $s \in \mathcal{S}$, $f, h \in C$, and $\alpha, \beta > 0$, define

$$g^s_{f,h,\alpha,\beta}(a, r) = \alpha f\big(u_s(a)\big) + \beta h\big(u_s(r)\big).$$

The local evaluation is separable in burden and reach. Separability expresses neutrality about burden–reach association: the evaluation depends on the anonymous burden and reach distributions but not on whether high values of the two attributes occur at the same positions. Monotonicity treats additional burden or reach, measured on the relevant scale, as hierarchy enhancing. Convexity makes the evaluation responsive to concentration: holding the total on that scale fixed, shifting burden or reach toward positions that already have higher values weakly raises the evaluation. The positive weights α and β determine the relative importance of the two profiles. Allowing all such functions and all positive relative weights leaves these numerical judgments unrestricted.

Definition 17 (Robust Marginal Unanimity)**.** Fix n and $s \in \mathcal{S}$. A hierarchical preorder $\succcurlyeq$ on $\mathcal{H}_n$ satisfies *Robust Marginal Unanimity* at scale s if, for every $G, H \in \mathcal{H}_n$:

1. if
$$\sum_{v \in V(G)} g^s_{f,h,\alpha,\beta}\big(a_G(v), r_G(v)\big) \geq \sum_{v \in V(H)} g^s_{f,h,\alpha,\beta}\big(a_H(v), r_H(v)\big)$$
for every $f, h \in C$ and every $\alpha, \beta > 0$, then $G \succcurlyeq H$;

2. if all these inequalities hold and at least one is strict, then $G \succ H$.

The strict clause requires every admissible evaluation to rank G weakly above H and at least one to rank it strictly above H. It does not require unanimous strictness, since constant or locally flat functions may assign the same value to both hierarchies.

We use the standard characterization of weak submajorization (Marshall, Olkin, and Arnold, 2011, Theorem 4.B.2). For two finite vectors $\mathbf{x}$ and $\mathbf{y}$ of equal length, the cumulative sums of the decreasing rearrangement of $\mathbf{x}$ are weakly greater than those of $\mathbf{y}$ at every cutoff if and only if

$$\sum_i f(x_i) \geq \sum_i f(y_i)$$

for every nondecreasing convex function f.

Theorem 1 (Fixed-size unanimity)**.** *Fix n and $s \in \mathcal{S}$. For $G, H \in \mathcal{H}_n$, the following statements are equivalent:*

1. $G \succcurlyeq^s_{\mathrm{AR}} H$*;*

2. *for every* $f, h \in C$ *and every* $\alpha, \beta > 0$,

$$\sum_{v \in V(G)} g^s_{f,h,\alpha,\beta}\big(a_G(v), r_G(v)\big) \geq \sum_{v \in V(H)} g^s_{f,h,\alpha,\beta}\big(a_H(v), r_H(v)\big).$$

Proof. If $G \succcurlyeq^s_{\mathrm{AR}} H$, the preceding characterization, applied separately to the burden and reach profiles measured on the relevant scale, gives

$$\sum_{v \in V(G)} f\big(u_s(a_G(v))\big) \geq \sum_{v \in V(H)} f\big(u_s(a_H(v))\big)$$

and

$$\sum_{v \in V(G)} h\big(u_s(r_G(v))\big) \geq \sum_{v \in V(H)} h\big(u_s(r_H(v))\big)$$

for every $f, h \in C$. Multiplying the first inequality by $\alpha > 0$, multiplying the second by $\beta > 0$, and adding proves statement 2.

Conversely, suppose the burden comparison fails. The same characterization provides $f \in C$ for which

$$\sum_{v \in V(G)} f\big(u_s(a_G(v))\big) < \sum_{v \in V(H)} f\big(u_s(a_H(v))\big).$$

Taking h to be the zero function, which belongs to C, contradicts statement 2. Hence the burden comparison must hold.

If the reach comparison fails, the analogous argument applies with f equal to the zero function and with a nondecreasing convex function h that detects the failure. Thus both marginal comparisons hold, and $G \succcurlyeq^s_{\mathrm{AR}} H$. ∎

Thus each fixed-size product preorder is exactly the unanimous relation generated by the stated marginal family. Individual evaluations may trade off burden against reach, but the product preorder retains only comparisons that survive every admissible marginal evaluation and every positive relative weight.

Corollary 1 (Robust-unanimity characterization)**.** *Fix n and $s \in \mathcal{S}$. A hierarchical preorder $\succcurlyeq$ on $\mathcal{H}_n$ satisfies Robust Marginal Unanimity at scale s if and only if it preserves every weak and strict comparison made by $\succcurlyeq^s_{\mathrm{AR}}$. That is,*

$$G \succcurlyeq^s_{\mathrm{AR}} H \Rightarrow G \succcurlyeq H$$

and

$$G \succ^s_{\mathrm{AR}} H \Rightarrow G \succ H.$$

Proof. Suppose $\succcurlyeq$ satisfies Robust Marginal Unanimity. If $G \succcurlyeq^s_{\mathrm{AR}} H$, every admissible evaluation weakly ranks G above H by Theorem 1, so $G \succcurlyeq H$.

If $G \succ^s_{\mathrm{AR}} H$, every admissible evaluation again weakly ranks G above H, and at least one must rank it strictly above H. Otherwise, every admissible evaluation would assign the same value to G and H, and Theorem 1 would imply $H \succcurlyeq^s_{\mathrm{AR}} G$, contradicting $G \succ^s_{\mathrm{AR}} H$. Therefore, the strict clause of Robust Marginal Unanimity gives $G \succ H$.

Conversely, suppose $\succcurlyeq$ preserves every weak and strict product comparison. If every admissible evaluation weakly ranks G above H, Theorem 1 gives $G \succcurlyeq^s_{\mathrm{AR}} H$, and hence $G \succcurlyeq H$.

If every admissible evaluation weakly ranks G above H and at least one ranks it strictly above H, then the reverse product comparison cannot hold. Thus $G \succ^s_{\mathrm{AR}} H$, and preservation of strict comparisons gives $G \succ H$. Both clauses of Robust Marginal Unanimity follow. ∎

The corollary identifies the normative content of the representation. A preorder satisfying Robust Marginal Unanimity may rank some pairs left incomparable by the product preorder, but it cannot reverse or weaken any comparison unanimously supported by the admissible marginal family.

11.2 Intensive and extensive unanimity

For the intensive interpretation, define

$$I^{s,I}_{f,h,\alpha,\beta}(G) = \frac{1}{|V(G)|} \sum_{v \in V(G)} g^{s}_{f,h,\alpha,\beta}\big(a_G(v), r_G(v)\big).$$

Corollary 2 (Intensive unanimity)**.** *For $s \in \mathcal{S}$ and $G, H \in \mathcal{H}$, $G \succcurlyeq^{s,I}_{\mathrm{AR}} H$ if and only if*

$$I^{s,I}_{f,h,\alpha,\beta}(G) \geq I^{s,I}_{f,h,\alpha,\beta}(H)$$

for every $f, h \in C$ and every $\alpha, \beta > 0$.

Proof. Write $|V(G)| = n$ and $|V(H)| = m$. By Definition 13 and Theorem 1, $G \succcurlyeq^{s,I}_{\mathrm{AR}} H$ holds if and only if every admissible total index ranks G^m above H^n. Dividing by the common population nm, or equivalently using replication invariance of the average indices, gives the stated equivalence. ∎

For the extensive interpretation, let

$$C_0 = \{f \in C : f(0) = 0\},$$

and, for $f, h \in C_0$, define

$$I^{s,E}_{f,h,\alpha,\beta}(G) = \sum_{v \in V(G)} g^{s}_{f,h,\alpha,\beta}\big(a_G(v), r_G(v)\big).$$

Since $u_s(0) = 0$ and $f(0) = h(0) = 0$, isolated positions contribute zero.

Corollary 3 (Extensive unanimity)**.** *For $s \in \mathcal{S}$ and $G, H \in \mathcal{H}$, $G \succcurlyeq^{s,E}_{\mathrm{AR}} H$ if and only if*

$$I^{s,E}_{f,h,\alpha,\beta}(G) \geq I^{s,E}_{f,h,\alpha,\beta}(H)$$

for every $f, h \in C_0$ and every $\alpha, \beta > 0$.

Proof. Let $N = \max\{|V(G)|, |V(H)|\}$ and zero-extend both hierarchies to population N. By Definition 14, the extensive comparison is the fixed-size comparison of G^{+N} and H^{+N}. Since functions in C_0 vanish at zero, adjoining isolated positions does not change the stated indices.

Although $C_0 \subseteq C$, restricting the fixed-size unanimity family to C_0 loses no comparisons. For any $f \in C$, define $\widetilde{f}(x) = f(x) - f(0)$. Then $\widetilde{f} \in C_0$. If $\mathbf{x}$ and $\mathbf{y}$ are two profiles of the common length N, then

$$\sum_{i=1}^{N} \widetilde{f}(x_i) = \sum_{i=1}^{N} f(x_i) - Nf(0) \quad \text{and} \quad \sum_{i=1}^{N} \widetilde{f}(y_i) = \sum_{i=1}^{N} f(y_i) - Nf(0).$$

Hence

$$\sum_{i=1}^{N} \widetilde{f}(x_i) \geq \sum_{i=1}^{N} \widetilde{f}(y_i) \Leftrightarrow \sum_{i=1}^{N} f(x_i) \geq \sum_{i=1}^{N} f(y_i).$$

The same argument applies to h. Thus every comparison detected by functions in C is also detected by functions in C_0, while the reverse inclusion is immediate because $C_0 \subseteq C$.

Therefore, unanimity over C_0 yields exactly the same fixed-size comparison as unanimity over C. The result follows from Theorem 1. ∎

These corollaries give the average and total representations associated with the intensive and extensive interpretations.

11.3 Concentration and normalized local positions

For the selected normalized-local concentration extension, define

$$I^{s,C}_{f,h,\alpha,\beta}(G) = \frac{1}{n} \sum_{v \in V(G)} \left[\alpha f\left(\widehat{a}^s_G(v)\right) + \beta h\left(\widehat{r}^s_G(v)\right)\right],$$

where $n = |V(G)|$, $f, h \in C$, and $\alpha, \beta > 0$. These indices evaluate the empirical distributions of normalized burden and reach and are invariant to common repetition of every normalized coordinate.

Proposition 14 (Concentration unanimity)**.** *For $s \in \mathcal{S}$ and $G, H \in \mathcal{H}$, $G \succcurlyeq^{s,C}_{\mathrm{AR}} H$ if and only if*

$$I^{s,C}_{f,h,\alpha,\beta}(G) \geq I^{s,C}_{f,h,\alpha,\beta}(H)$$

for every $f, h \in C$ and every $\alpha, \beta > 0$.

Proof. Write $|V(G)| = n$ and $|V(H)| = m$. Repeat each normalized coordinate of G m times and each normalized coordinate of H n times. The resulting profiles have length nm.

The nondecreasing-convex characterization applied to the repeated normalized burden profiles gives

$$m \sum_{v \in V(G)} f\left(\widehat{a}^s_G(v)\right) \geq n \sum_{v \in V(H)} f\left(\widehat{a}^s_H(v)\right)$$

for every $f \in C$ exactly when the normalized burden comparison holds. Dividing by nm gives the corresponding average inequality. The same argument applies to normalized reach. Multiplication by $\alpha, \beta > 0$ and addition prove the forward implication.

Conversely, if either normalized profile comparison fails, the majorization characterization provides a nondecreasing convex function detecting the failure. Taking the other marginal function to be zero contradicts the assumed index inequalities. Hence both normalized profile comparisons hold. ∎

This family differs from the pair-normalized concentration indices studied by Carbonell-Nicolau (2026a), which take the form

$$\frac{1}{\binom{n}{2}} \sum_{v \in V(G)} g\left(a_G(v), r_G(v)\right), \qquad n \geq 2. \tag{2}$$

Those indices assign each hierarchy a scalar value. They apply a population-independent local function to the raw burden and reach counts, sum the resulting position-level scores, and divide the total by the number of potential authority relations.

The present construction serves a different purpose. The fixed-size product preorder compares the complete burden and reach profiles at every rank cutoff, so its concentration extension requires position-level profiles that are comparable across population sizes. For this reason, each transformed burden and reach coordinate is divided by its largest feasible value at the relevant population size. The resulting normalized profiles describe each position's authority and subjection relative to what is possible in an organization of that size. Repetition then places these profiles on a common population grid for cumulative comparison.

The numerical representation averages evaluations of the normalized coordinates:

$$\frac{1}{n} \sum_{v \in V(G)} \left[\alpha f\left(\widehat{a}^s_G(v)\right) + \beta h\left(\widehat{r}^s_G(v)\right)\right].$$

The factor $1/n$ makes the representation depend on the empirical distributions of the normalized coordinates rather than on the number of positions. In particular, repeating every normalized coordinate leaves the

empirical distributions, and hence their average evaluations, unchanged, as required by the population-share comparison underlying the concentration preorder.

For linear marginal functions under the absolute scale, the pair-normalized indices in (2) and the normalized-local evaluations used here differ only by a constant factor. For nonlinear marginal functions or under the proportional scale, evaluating population-normalized coordinates need not agree with evaluating raw coordinates and then normalizing the aggregate.

11.4 Marginal sufficiency and the maximal local class

The preceding results show that separable evaluations with nondecreasing convex marginal components recover the product preorders by unanimity. We now ask whether this family exhausts the compatible position-additive indices. For absolute dominance, the answer is affirmative up to index-equivalent representations, provided that one population-independent local evaluation preserves all absolute product comparisons across finite population sizes.

A general local evaluation $g(a, r)$ can respond to the association between burden and reach across positions. For example, a term such as ar assigns additional significance to positions that simultaneously have positive burden and positive reach. Such positions may be viewed as intermediaries, coordinators, or organizational bottlenecks. Other nonseparable evaluations could instead favor a sharper separation between positions that exercise authority and positions that are subject to it, treating negative burden–reach association as an additional form of hierarchical role polarization.

The direction of either judgment depends on whether burden and reach at the same position are viewed as complements or substitutes. The product preorder remains neutral about this association. The padding construction below shows that any population-independent local evaluation preserving every absolute product comparison across finite population sizes must share this neutrality and be additively separable in burden and reach.

For $a, r \in \mathbb{Z}_+$, let $C(a, r)$ be the three-level hierarchy with a upper positions, one focal position, and r lower positions. Each upper position points to the focal position, which points to each lower position. The focal position has burden a and reach r. Each upper position has burden 0 and reach $r + 1$, while each lower position has burden $a + 1$ and reach 0. Thus the hierarchy contains one position with burden–reach pair (a, r), a positions with pair $(0, r + 1)$, and r positions with pair $(a + 1, 0)$.

Let O_t be the out-star with t leaves. Its root has burden 0 and reach t, while each leaf has burden 1 and reach 0. Thus O_t contains one position with burden–reach pair $(0, t)$ and t positions with pair $(1, 0)$.

Let I_t be the in-star with t roots and one common leaf. The leaf has burden t and reach 0, while each root has burden 0 and reach 1. Thus I_t contains one position with burden–reach pair $(t, 0)$ and t positions with pair $(0, 1)$.

To compare the numbers of positions with different burden–reach pairs, we use formal symbols $\mathbf{e}_{(a,r)}$, one for each $(a, r) \in \mathbb{Z}_+^2$. The coefficient of $\mathbf{e}_{(a,r)}$ records the difference between the numbers of positions with burden a and reach r in the two hierarchies being compared. Thus, for example,

$$2\mathbf{e}_{(a,r)} - \mathbf{e}_{(a',r')}$$

means that the first hierarchy has two more positions with pair (a, r) and one fewer position with pair (a', r') than the second.

Lemma 4 (Rectangular identities across population sizes)**.** *Let $g : \mathbb{Z}_+^2 \to \mathbb{R}$ be population independent, and define*

$$I_g(G) = \sum_{v \in V(G)} g\big(a_G(v), r_G(v)\big).$$

Suppose I_g preserves $\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}}$ on every $\mathcal{H}_n$. Then, for all $a < a'$ and $r < r'$,

$$g(a,r) + g(a',r') = g(a,r') + g(a',r).$$

Consequently, there exist functions $F, Q : \mathbb{Z}_+ \to \mathbb{R}$ and a constant c such that

$$g(a,r) = F(a) + Q(r) + c$$

for every $(a,r) \in \mathbb{Z}_+^2$.

Proof. Fix $a < a'$ and $r < r'$, and write $d = a' - a$ and $e = r' - r$. Begin with

$$G_0 = \big(C(a,r), C(a',r')\big) \quad \text{and} \quad H_0 = \big(C(a,r'), C(a',r)\big).$$

The two focal positions in G_0 have burden–reach pairs (a,r) and (a',r'), while those in H_0 have pairs (a,r') and (a',r). Thus the contribution of the focal positions to the multiplicity difference "G_0 minus H_0" is

$$\mathbf{e}_{(a,r)} + \mathbf{e}_{(a',r')} - \mathbf{e}_{(a,r')} - \mathbf{e}_{(a',r)}.$$

Consider next the nonfocal positions. Among the upper positions, G_0 has a positions with burden–reach pair $(0, r+1)$ and a' positions with pair $(0, r'+1)$. In contrast, H_0 has a' positions with pair $(0, r+1)$ and a positions with pair $(0, r'+1)$. Since $d = a' - a$, the contribution of the upper positions to the multiplicity difference "G_0 minus H_0" is

$$d\big[\mathbf{e}_{(0,r'+1)} - \mathbf{e}_{(0,r+1)}\big].$$

Among the lower positions, G_0 has r positions with pair $(a+1, 0)$ and r' positions with pair $(a'+1, 0)$, whereas H_0 has r' positions with pair $(a+1, 0)$ and r positions with pair $(a'+1, 0)$. Since $e = r' - r$, the contribution of the lower positions to the same multiplicity difference is

$$e\big[\mathbf{e}_{(a'+1,0)} - \mathbf{e}_{(a+1,0)}\big].$$

We now add position-disjoint components to cancel the multiplicity differences contributed by the nonfocal positions.

First, adjoin d copies of O_{r+1} to G_0 and d copies of $O_{r'+1}$ to H_0. Each copy of O_t contains one root with burden–reach pair $(0,t)$ and t leaves with pair $(1,0)$. The contribution of this padding to the multiplicity difference "G_0 minus H_0" is therefore

$$d\big[\mathbf{e}_{(0,r+1)} - \mathbf{e}_{(0,r'+1)}\big] + d\big[(r+1) - (r'+1)\big]\mathbf{e}_{(1,0)}.$$

Since $e = r' - r$, this expression equals

$$d\big[\mathbf{e}_{(0,r+1)} - \mathbf{e}_{(0,r'+1)}\big] - de\,\mathbf{e}_{(1,0)}.$$

Its first term cancels the multiplicity difference previously contributed by the upper positions. The remaining difference is

$$-de\,\mathbf{e}_{(1,0)}.$$

Next, adjoin e copies of I_{a+1} to G_0 and e copies of $I_{a'+1}$ to H_0. Each copy of I_t contains one leaf with burden–reach pair $(t,0)$ and t roots with pair $(0,1)$. The contribution of this padding is

$$e\big[\mathbf{e}_{(a+1,0)} - \mathbf{e}_{(a'+1,0)}\big] + e\big[(a+1) - (a'+1)\big]\mathbf{e}_{(0,1)}.$$

Since $d = a' - a$, this expression equals

$$e\left[\mathbf{e}_{(a+1,0)} - \mathbf{e}_{(a'+1,0)}\right] - de\,\mathbf{e}_{(0,1)}.$$

Its first term cancels the multiplicity difference contributed by the lower positions. The remaining difference is

$$-de\,\mathbf{e}_{(0,1)}.$$

At this stage, apart from the focal positions, the multiplicity difference is

$$-de\,\mathbf{e}_{(1,0)} - de\,\mathbf{e}_{(0,1)}.$$

Finally, adjoin de single-edge components to G_0. Each such component contains one position with pair $(0,1)$ and one with pair $(1,0)$. Their combined contribution is

$$de\,\mathbf{e}_{(1,0)} + de\,\mathbf{e}_{(0,1)},$$

which cancels both remaining differences.

Call the resulting hierarchies G and H. We first verify that they have the same population. Relative to the padding added to H, the out-star padding added to G contains de fewer positions: each of the d out-stars added to G has $e = r' - r$ fewer leaves. Similarly, the in-star padding added to G contains de fewer positions than the in-star padding added to H: each of the e in-stars added to G has $d = a' - a$ fewer roots. The de single-edge components added to G contribute $2de$ positions. These differences sum to zero, so $|V(G)| = |V(H)|$.

The padding cancels every burden–reach multiplicity difference except that of the four focal positions. Thus the remaining multiplicity difference "G minus H" is

$$\mathbf{e}_{(a,r)} + \mathbf{e}_{(a',r')} - \mathbf{e}_{(a,r')} - \mathbf{e}_{(a',r)}.$$

The two positive pairs and the two negative pairs have the same burden multiset, $\{a, a'\}$, and the same reach multiset, $\{r, r'\}$. Consequently, G and H have identical anonymous burden profiles and identical anonymous reach profiles. Hence $G \sim^{\text{abs}}_{\text{AR}} H$.

Since $G \sim^{\text{abs}}_{\text{AR}} H$ and I_g preserves every absolute product comparison, preservation in both directions gives

$$I_g(G) \geq I_g(H) \quad \text{and} \quad I_g(H) \geq I_g(G).$$

Hence $I_g(G) = I_g(H)$.

By construction, G and H contain the same number of positions with every burden–reach pair except the four focal pairs. Relative to H, the hierarchy G has one additional position with pair (a, r) and one with pair (a', r'), and one fewer position with pair (a, r') and one with pair (a', r). Therefore,

$$I_g(G) - I_g(H) = g(a,r) + g(a',r') - g(a,r') - g(a',r).$$

Since $I_g(G) = I_g(H)$, it follows that

$$g(a,r) + g(a',r') = g(a,r') + g(a',r).$$

For $a, r > 0$, apply this rectangular identity with the four pairs $(0,0)$, $(a,0)$, $(0,r)$, and (a,r). It gives

$$g(0,0) + g(a,r) = g(0,r) + g(a,0),$$

and therefore

$$g(a,r) = g(a,0) + g(0,r) - g(0,0).$$

Define

$$F(a) = g(a,0) - g(0,0), \qquad Q(r) = g(0,r) - g(0,0), \qquad c = g(0,0).$$

Then

$$g(a,r) = F(a) + Q(r) + c.$$

If $a = 0$, then $F(0) = 0$, and the definitions give

$$F(0) + Q(r) + c = g(0,r).$$

Similarly, if $r = 0$, then $Q(0) = 0$, and

$$F(a) + Q(0) + c = g(a,0).$$

Thus the decomposition holds for every $(a,r) \in \mathbb{Z}_+^2$. ∎

The construction realizes every finite rectangle of burden–reach pairs within a pair of sufficiently large DAGs. Position-disjoint components are added so that the two hierarchies have the same population and the same numbers of positions at every burden–reach pair except the four corners of the rectangle. At those four corners, the hierarchies have the same anonymous burden profile and the same anonymous reach profile, but differ in how burden and reach are associated across positions. Preservation in both directions forces the rectangular identity.

The additional components allow this construction to be carried out at a sufficiently large population size; the relevant rearrangement need not be feasible within any particular fixed-size domain. Therefore, the requirement that one local evaluation preserve absolute product comparisons across all finite population sizes is essential.

The rectangular identities establish additive separability of every compatible local evaluation within the all-size architecture. It remains to determine the restrictions that product preservation imposes on the two marginal components. The next lemma shows that each component must be discretely convex.

Lemma 5 (Component convexity)**.** *Suppose $g(a,r) = F(a) + Q(r) + c$ and I_g preserves $\succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}}$ on every $\mathcal{H}_n$. Then, for every $k \geq 1$,*

$$F(k+1) + F(k-1) \geq 2F(k) \quad \textit{and} \quad Q(k+1) + Q(k-1) \geq 2Q(k).$$

Proof. Fix $k \geq 1$. Construct two two-level bipartite hierarchies G_k and H_k, each with upper positions $1, \ldots, k+1$ and lower positions x and y.

In G_k, every upper position supervises x, while positions $1, \ldots, k-1$ also supervise y. Thus

$$E(G_k) = \{i \to x : i = 1, \ldots, k+1\} \cup \{i \to y : i = 1, \ldots, k-1\}.$$

The burdens of x and y are $k+1$ and $k-1$, respectively. Positions $1, \ldots, k-1$ each reach both lower positions, while positions k and $k+1$ reach only x. Hence the ordered reach profile of the upper positions is

$$(\underbrace{2, \ldots, 2}_{k-1 \text{ copies}}, 1, 1).$$

In H_k, positions $1, \ldots, k$ supervise x, while positions $1, \ldots, k-1$ and $k+1$ supervise y. Thus

$$E(H_k) = \{i \to x : i = 1, \ldots, k\} \cup \{i \to y : i = 1, \ldots, k-1\} \cup \{k+1 \to y\}.$$

Both lower positions have burden k. Positions $1, \dots, k-1$ again reach both lower positions, while positions k and $k+1$ each reach one. Hence the ordered reach profile of the upper positions is again

$$(\underbrace{2, \dots, 2}_{k-1 \text{ copies}}, 1, 1).$$

The complete reach profiles of G_k and H_k coincide because the two lower positions have reach zero in both hierarchies. Their burden profiles differ only at the lower positions:

$$(k+1, k-1) \quad \text{in } G_k, \qquad (k, k) \quad \text{in } H_k.$$

Since $(k+1, k-1)$ majorizes (k, k),

$$G_k \succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} H_k.$$

Preservation gives $I_g(G_k) \geq I_g(H_k)$. Because $g(a, r) = F(a) + Q(r) + c$, the identical reach profiles make the Q-contributions equal. The upper positions have burden zero in both hierarchies, so their F-contributions are also equal. The populations coincide, so the constant contributions cancel. The remaining inequality is

$$F(k+1) + F(k-1) \geq 2F(k).$$

Reversing every edge exchanges burden and reach. Applying the same argument to the reversed hierarchies gives

$$Q(k+1) + Q(k-1) \geq 2Q(k). \qquad \blacksquare$$

Product preservation alone does not force a particular separable representation to have nondecreasing components. For example, let $g(a, r) = -a + 2r$. The common-volume identity gives

$$I_g^E(G) = -\sum_{v \in V(G)} a_G(v) + 2 \sum_{v \in V(G)} r_G(v) = |P_G|,$$

even though the burden component is decreasing.

This lack of monotonicity reflects an indeterminacy in the separable representation rather than in the resulting hierarchy index. Suppose

$$g(a, r) = F(a) + Q(r) + c.$$

For any $t \in \mathbb{R}$, define

$$F_t(a) = F(a) + ta \quad \text{and} \quad Q_t(r) = Q(r) - tr.$$

The corresponding local evaluation satisfies

$$g_t(a, r) = g(a, r) + t(a - r).$$

By the common-volume identity,

$$I_{g_t}^E(G) - I_g^E(G) = t\left(\sum_{v \in V(G)} a_G(v) - \sum_{v \in V(G)} r_G(v)\right) = 0.$$

Thus g_t and g induce the same index on every hierarchy.

If I_g^E preserves all absolute product comparisons, comparison of a single-edge hierarchy with the empty hierarchy gives

$$F(1) - F(0) + Q(1) - Q(0) \geq 0.$$

Since F and Q are discretely convex, their successive differences are nondecreasing. Hence one may choose t such that

$$-\big(F(1)-F(0)\big) \le t \le Q(1)-Q(0).$$

For this choice, both F_t and Q_t are nondecreasing. Therefore, every preserving index admits a representation with nondecreasing, discretely convex marginal components, even though the original local evaluation need not itself be coordinatewise nondecreasing.

Theorem 2 (Maximal absolute local-index class)**.** *Let $g : \mathbb{Z}_+^2 \to \mathbb{R}$ be population independent, and define*

$$I_g^E(G) = \sum_{v \in V(G)} g\big(a_G(v), r_G(v)\big) \quad \text{and} \quad I_g^I(G) = \frac{1}{|V(G)|} \sum_{v \in V(G)} g\big(a_G(v), r_G(v)\big).$$

The following statements are equivalent:

1. *for every n and every $G, H \in \mathcal{H}_n$,*

$$G \succcurlyeq_{\mathrm{AR}}^{\mathrm{abs}} H \Rightarrow I_g^E(G) \ge I_g^E(H);$$

2. *I_g^I preserves absolute intensive dominance on $\mathcal{H}$;*

3. *there exist nondecreasing, discretely convex functions $\overline{F}, \overline{Q} : \mathbb{Z}_+ \to \mathbb{R}$ and a constant c such that the local evaluation*

$$\overline{g}(a, r) = \overline{F}(a) + \overline{Q}(r) + c$$

induces the same index as g on every hierarchy:

$$I_{\overline{g}}^E(G) = I_g^E(G)$$

for every $G \in \mathcal{H}$.

If $g(0,0) = 0$, these statements are also equivalent to:

4. *I_g^E preserves absolute extensive dominance on $\mathcal{H}$.*

Proof. (1) $\Rightarrow$ (3). By Lemma 4, statement 1 implies that

$$g(a, r) = F(a) + Q(r) + c$$

for some functions $F, Q : \mathbb{Z}_+ \to \mathbb{R}$ and a constant c. By Lemma 5, both F and Q are discretely convex.

We next show that the index admits a representation with nondecreasing components. Define

$$\Delta F(k) = F(k+1) - F(k) \quad \text{and} \quad \Delta Q(k) = Q(k+1) - Q(k).$$

Discrete convexity implies

$$\Delta F(k) \ge \Delta F(0) \quad \text{and} \quad \Delta Q(k) \ge \Delta Q(0)$$

for every $k \ge 0$.

Compare the two-position single-edge hierarchy with the two-position empty hierarchy. Statement 1 gives

$$\Delta F(0) + \Delta Q(0) \ge 0.$$

Thus the interval

$$\big[-\Delta F(0), \Delta Q(0)\big]$$

is nonempty. Choose any t in this interval and define

$$\overline{F}(a) = F(a) + ta, \qquad \overline{Q}(r) = Q(r) - tr.$$

Both functions remain discretely convex. Moreover,

$$\overline{F}(k+1) - \overline{F}(k) = \Delta F(k) + t \geq \Delta F(0) + t \geq 0$$

and

$$\overline{Q}(k+1) - \overline{Q}(k) = \Delta Q(k) - t \geq \Delta Q(0) - t \geq 0.$$

Hence $\overline{F}$ and $\overline{Q}$ are nondecreasing.

Let

$$\overline{g}(a, r) = \overline{F}(a) + \overline{Q}(r) + c.$$

For every hierarchy G, Lemma 1 gives

$$\sum_{v \in V(G)} a_G(v) = \sum_{v \in V(G)} r_G(v).$$

Therefore,

$$I^E_{\overline{g}}(G) - I^E_g(G) = t \sum_{v \in V(G)} \big(a_G(v) - r_G(v)\big) = t\left(\sum_{v \in V(G)} a_G(v) - \sum_{v \in V(G)} r_G(v)\right) = 0.$$

Hence $I^E_{\overline{g}}(G) = I^E_g(G)$ for every hierarchy G, proving statement 3.

(3) $\Rightarrow$ (1). Suppose $G \succcurlyeq^{\text{abs}}_{\text{AR}} H$. Since $\overline{F}$ and $\overline{Q}$ are nondecreasing and discretely convex, weak majorization gives

$$\sum_{v \in V(G)} \overline{F}(a_G(v)) \geq \sum_{v \in V(H)} \overline{F}(a_H(v)) \quad \text{and} \quad \sum_{v \in V(G)} \overline{Q}(r_G(v)) \geq \sum_{v \in V(H)} \overline{Q}(r_H(v)).$$

Because $G, H \in \mathcal{H}_n$, adding the two marginal inequalities and the common term nc gives $I^E_{\overline{g}}(G) \geq I^E_{\overline{g}}(H)$. Since $I^E_{\overline{g}} = I^E_g$ on every hierarchy, $I^E_g(G) \geq I^E_g(H)$.

(2) $\Rightarrow$ (1) follows from fixed-size agreement and $I^E_g(G) = nI^I_g(G)$ on $\mathcal{H}_n$.

(1) $\Rightarrow$ (2). If $|V(G)| = n$, $|V(H)| = m$, and $G \succcurlyeq^{\text{abs},I}_{\text{AR}} H$, then $G^m \succcurlyeq^{\text{abs}}_{\text{AR}} H^n$. Statement 1 and

$$I^E_g(G^m) = mI^E_g(G), \qquad I^E_g(H^n) = nI^E_g(H),$$

imply $I^I_g(G) \geq I^I_g(H)$.

(4) $\Rightarrow$ (1) follows by restriction to a fixed-size domain. Conversely, assume statement 1 and $g(0,0) = 0$. Zero-extending the smaller hierarchy leaves I^E_g unchanged and reduces every extensive comparison to a fixed-size comparison. Hence statement 1 implies statement 4. ∎

The all-size requirement is essential. One common local evaluation must preserve the absolute product comparisons at every finite population size. The theorem does not characterize the compatible position-additive indices on an isolated fixed-size domain. At a fixed population size, the restricted set of feasible burden–reach pairs may permit nonseparable or otherwise size-specific evaluations that do not preserve product comparisons across all finite sizes.

Together, Theorems 1 and 2 show that the absolute marginal family is both separating and exhaustive, up to index-equivalent representations, within the stated all-size framework. The unanimity theorem shows

that the family recovers every absolute product comparison. The maximality theorem shows that every population-independent position-additive index preserving all such comparisons across finite population sizes admits an equivalent representation with nondecreasing, discretely convex burden and reach components.

A nonseparable index may distinguish pairs left unresolved by the product preorder, but it does so by introducing a judgment about burden–reach association. It may assign additional significance to intermediary positions where burden and reach are both high or, conversely, favor a sharper separation between positions that exercise authority and positions that are subject to it. The product preorder does not reject either interpretation; it retains only comparisons that do not depend on choosing between them.

The fixed-size unanimity representation applies to both scales, but the maximality theorem is absolute. The rectangular argument continues to force separability under proportional dominance because it uses only product equivalence. The convexity construction does not transfer: although $(k+1, k-1)$ majorizes (k, k) in raw counts,

$$\log(k+2) + \log k < 2\log(k+1),$$

so the transformed profiles cross. A proportional maximality theorem would require multiplicative transfer constructions adapted to the logarithmic grid.

A concentration maximality theorem poses a different problem because the local domain and normalization denominator change with population. Compatibility would have to be stated for a family $(g_n)_{n\geq 2}$ linked across sizes by an additional normalized-locality principle. We leave the proportional and concentration characterizations for separate analysis.

12 Scope and alternative interpretations

The absolute and proportional relations measure related but distinct forms of two-sided hierarchical stratification. Absolute dominance compares effective authority relations in their original units, whereas proportional dominance compares proportional changes in inclusive burden and reach. Neither relation is intended to capture every organizational feature associated with hierarchy.

The burden and reach profiles are the two anonymous margins of the reachability relation, not a complete description of it. Burden does not distinguish whether a position's supervisors lie along one chain or on several converging branches, and reach does not distinguish the internal organization of a position's subordinate domain. The criterion also concerns effective reachability rather than direct reporting lines. It is insensitive to redundant edges and does not separately measure direct span of control, unity of command, or reporting-link density.

The product preorders compare concentration within the burden and reach profiles separately but do not compare the association between the two attributes across positions. Hierarchies with the same anonymous burden and reach profiles are equivalent even if high burden and high reach occur at the same positions in one hierarchy and at different positions in the other. Distinguishing such hierarchies requires an additional judgment about the organizational significance of positions that both bear substantial supervisory burden and exercise substantial subordinate reach.

One possible refinement would regard stronger negative association between burden and reach as an additional form of hierarchical role polarization. Under this interpretation, hierarchy is more pronounced when positions with high reach tend to have low burden and positions with high burden tend to have low reach. Such a refinement could distinguish hierarchies that the present product preorders treat as equivalent. These hierarchies would have the same anonymous burden and reach profiles but different associations between the two attributes across positions.

Stronger negative association need not, however, represent greater hierarchy in every application. Positions with both positive burden and positive reach may represent intermediate layers of command, coordination, or organizational bottlenecks. Their presence may be viewed as evidence of hierarchical intermediation

rather than as a departure from role polarization. Comparing burden–reach association requires an additional judgment about which organizational feature is relevant. The product preorders remain neutral about that judgment and use only the two anonymous profiles.

The preorders also remain neutral about the relative importance of burden and reach. Robust Marginal Unanimity retains only comparisons that survive every positive relative weight on the two profiles. The identity and logarithmic transformations likewise represent two focal attitudes toward scale: equal absolute changes and equal proportional changes in inclusive domains. Other transformations are mathematically possible but require their own organizational interpretation.

Alternative criteria may appropriately introduce judgments that the product preorders leave open. A weighted scalar rule selects a burden–reach tradeoff, a lexicographic rule gives one profile priority, and a nonseparable local evaluation responds to burden–reach association. An order on the full reachability relation can use structural information beyond the anonymous margins, while alternative profile transformations can express different attitudes toward scale.

The framework does not reject more discriminating criteria. It identifies the comparisons that do not depend on these additional judgments. When robust agreement fails, a more complete ranking requires an organizational basis for resolving the disagreement.

13 Conclusion

Every effective authority relation connects a supervisory position to a subordinate position. Reach and burden record the two anonymous margins of this relation and have the same aggregate total, but they describe different distributions: subordinate reach and supervisory burden. Absolute and proportional two-sided dominance rank one hierarchy above another only when the burden and reach comparisons support that ranking at every cutoff.

The tree–diamond example shows why both profiles matter. The diamond concentrates subjection, while the tree concentrates intermediate authority. The two one-sided comparisons give opposite rankings, and scalar rankings reverse as the relative burden and reach weights change. The product preorder records the resulting absence of robust agreement rather than selecting one perspective or burden–reach tradeoff.

Both product relations satisfy Subordination Removal and preserve every weak and strict fixed-size comparison generated by a sequence of such removals. The converse fails: cumulative burden and reach comparisons can rank hierarchies that cannot be related through a removal sequence. Flat hierarchies and total reachability orders provide the fixed-size lower and upper benchmarks.

Comparisons across population sizes require a separate judgment about how organizational size enters the evaluation. The per-position extension compares organizations after replication to a common population and is uniquely determined by fixed-size agreement and replication invariance. Vertical attachment raises hierarchy under this interpretation when the appended component is at least as hierarchical as the base, but need not do so without that condition.

The aggregate extension compares organizations after adjoining isolated positions to the smaller one. It is uniquely determined by fixed-size agreement and neutrality toward isolated positions, and every admissible vertical attachment raises aggregate hierarchy. The population-normalized extension compares each position's burden and reach with what would be possible in an organization of the same size. Under this interpretation, adding positions can lower measured hierarchy even when it creates new authority relations, because the range of possible authority relations expands as well.

The numerical results formalize the robust-unanimity interpretation. For either profile scale, the fixed-size product preorder is exactly the unanimous relation generated by separable evaluations with nondecreasing convex marginal components and freely varying positive burden and reach weights. The per-position, aggregate, and population-normalized extensions have corresponding average, total, and normalized-local

representations.

For absolute dominance, the maximality result is stronger. Consider a population-independent index formed by assigning each position a score based on its burden and reach and summing these scores across positions. If the same local evaluation preserves every absolute product comparison at every finite population size, the resulting index admits an equivalent representation that is additively separable in burden and reach, with nondecreasing, discretely convex components. The absolute marginal family is both separating and exhaustive, up to index-equivalent representations, within this all-size position-additive class.

The resulting framework is a robust but incomplete measure of two-sided hierarchical stratification. It distinguishes comparisons that are robust to the admissible numerical judgments from those that require additional judgments. It compares the anonymous burden and reach profiles while remaining neutral about their association across positions. More discriminating criteria may appropriately introduce additional judgments about concentration, burden–reach tradeoffs, priority, burden–reach association, or the internal structure of the reachability relation.

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